Phase 4 Final Report: SNAAK Cervical Biopsy Forceps

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BME 60C Spring 2026

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8 June 2026

Existing Research and Opportunities

Cervical cancer is the fourth most common type of cancer and affects countless women worldwide. In 2022, it was reported that approximately 660,000 women received a cervical cancer diagnosis and 350,000 died from the disease in that year alone. This highlights the importance of effective screening and early diagnosis. Early detection is heavily reliant on obtaining high-quality tissue samples from the cervix for pathological evaluation, which makes cervical biopsy extremely essential for women's healthcare. **[1]**

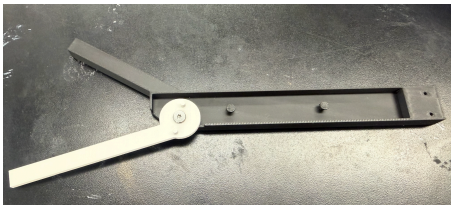
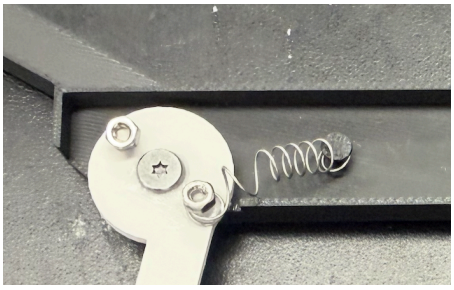
A cervical biopsy involves cutting and removing a small sample of tissue from the cervix to identify precancerous or cancerous cells. **[2]** Currently, forceps use opposing jaws to collect samples but often encounter challenges related to specimen retention and collection consistency. Tissue samples may become dislodged during withdrawal due to mucus, patient movement, or insufficient gripping force, resulting in inadequate specimens that can require repeat procedures. Additionally, biopsy depth is highly dependent on physician technique, which contributes to variability in sample quality and diagnostic reliability. **[3]**

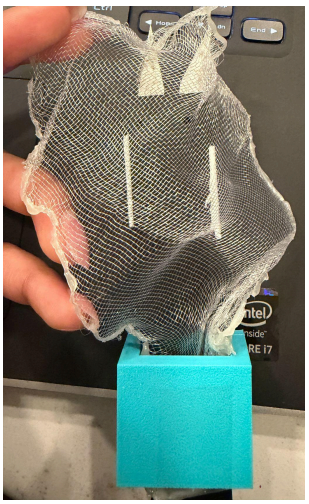
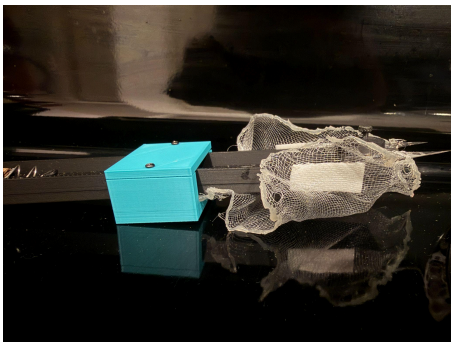


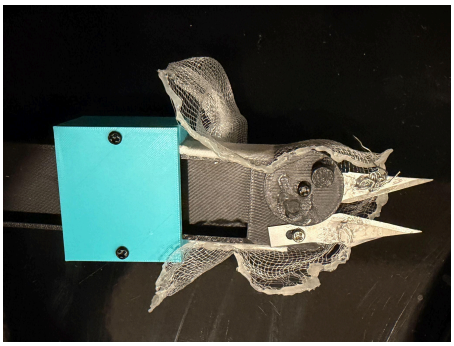
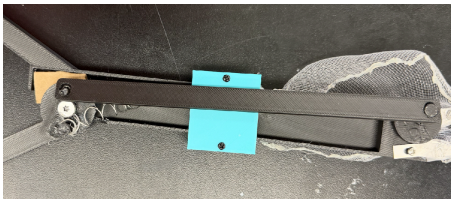
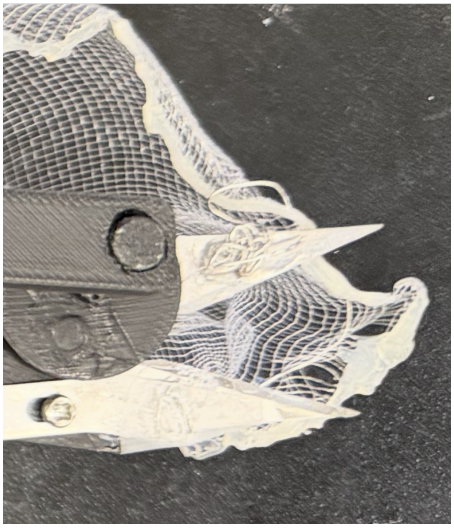
Figure 1. *Design of current biopsy forceps - an eagle-shaped jaw and three stop bars help with positioning and grabbing tissues. These devices use jaws to hold tissues, lack fully-secure retrieval, and do not account for bite depth* ***[4]****.*

The existing limitations create both clinical and economic disadvantages. Repeat biopsies increase risk of cervical bleeding, patient discomfort and anxiety, procedure times, and consume additional medical supplies. There is a need for a biopsy device that improves sample reliability while remaining simple, affordable, and compatible with existing procedures and technologies. To address these limitations, our team designed PetalGrip, an improved cervical biopsy forceps device, which includes a base, dual rotating turntables integrated into the handle assembly, a central linkage rod, a compression spring, and an integrated tissue retention net. The turntable-linkage mechanism converts the squeezing motion of the handles into controlled translational motion, improving force transmission and jaw actuation. Compression springs maintain consistent tension within the mechanism and assist in reliable opening and closing of the jaws. The primary innovation of our design is the integrated tissue retention net. Unlike current forceps that rely solely on jaw compression, this net provides an additional physical capture system to secure and retain the sample. Hence, this improves specimen retention, preserves sample integrity, and reduces repeat procedures. It enhances clinical efficiency and patient health and comfort.

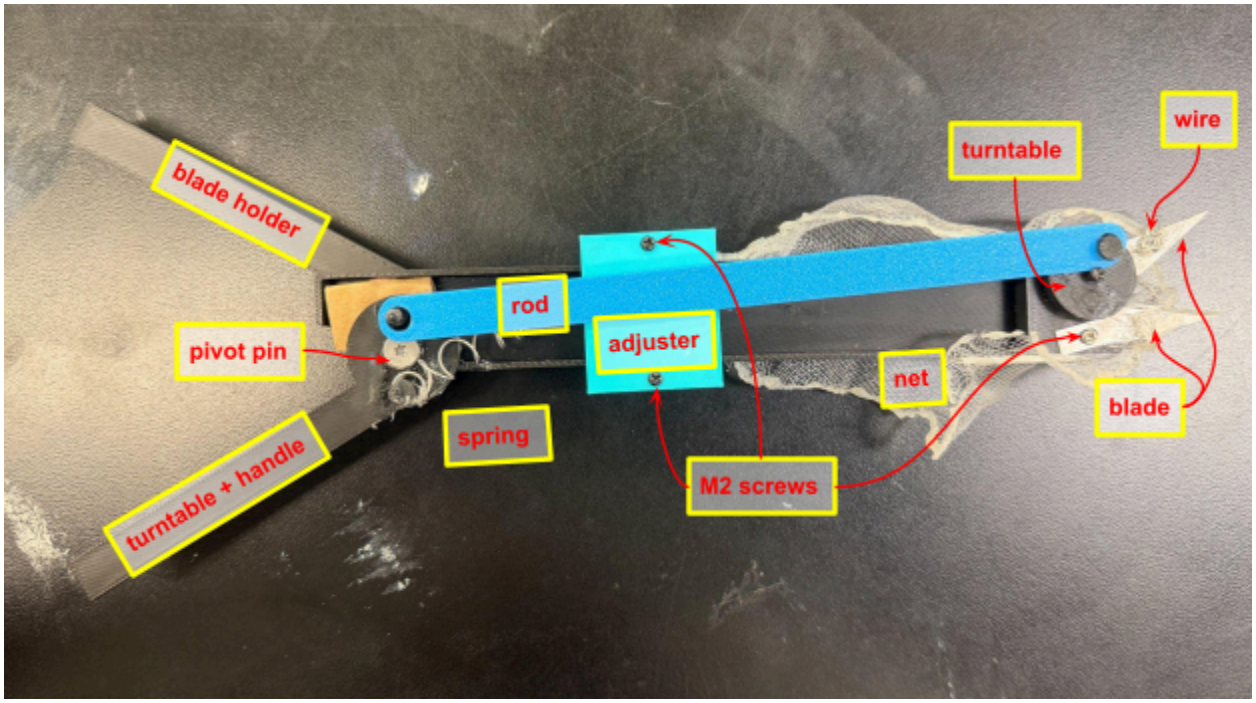
Prototype Design - Detailed Steps

Steps	Image of Model
Orient the blade holder so the handle/thumb-rest side is positioned toward the upper left.	
Attach the turntable handle to the blade holder using a 10 mm pivot pin.	
Attach the compression spring, 0.025 x 0.276 x 0.787 in / 0.65 x 7 x 20 mm, between the hook on the turntable handle that is closer to you and the hook on the left side of the blade holder. Secure both ends of the spring with super glue.	
Orient one blade with the point facing right and slightly angled upwards, and the sharp edge facing upward, away from you.	
Fasten this blade to the bottom hole of the blade holder using an M2 x 3 mm screw.	
Orient the second blade with the point facing right and the sharp edge facing downward, toward you.	
Apply super glue to the rectangular base of the second blade, then place the turntable on top of it. Ensure the hook faces upward and is positioned near the exposed blade edge.	
Loosely screw the turntable and second blade to the blade holder using an M2 x 3 mm screw.	

Cut an oval piece of mesh approximately 5 in long and 4 in wide.	
Use hot glue around the perimeter of the mesh to prevent fraying.	
Orient the net adjuster's base with the large flat side facing down and the two raised edges facing upward. From your view, the base should be in portrait orientation.	
Super glue one end of the mesh oval to the inner surface of the adjuster base.	
Place the middle section of the blade holder onto the adjuster base, with the excess mesh extending to the right side.	
Place the adjuster lid on top of the blade holder and align it with the adjuster base.	
Fasten the adjuster base and lid together using two M2 x 3 mm screws.	

Slide the adjuster fully to the right, toward the blades.	
Attach the connecting rod between the two turntables. Place one end of the rod around the hook on the handle (replaced with an M2 screw temporarily due to hook breakage), then place the other end around the hook on the turntable attached to the blade's base.	
Cut a piece of wire approximately 4 in long. Thread the wire through the mesh, then super glue both ends of the wire to the blades to connect the mesh to the blade mechanism.	

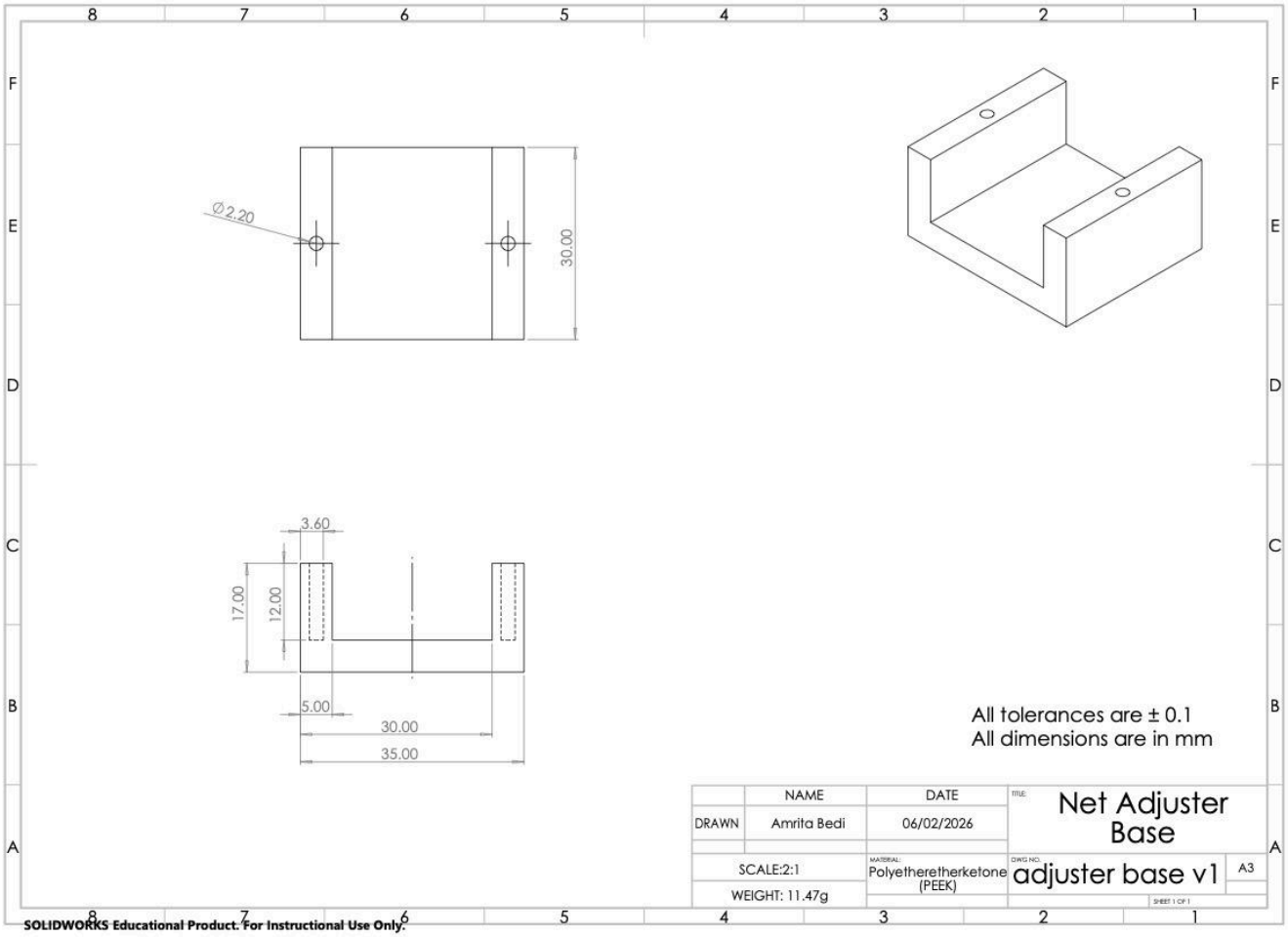
Final Prototype (labeled):



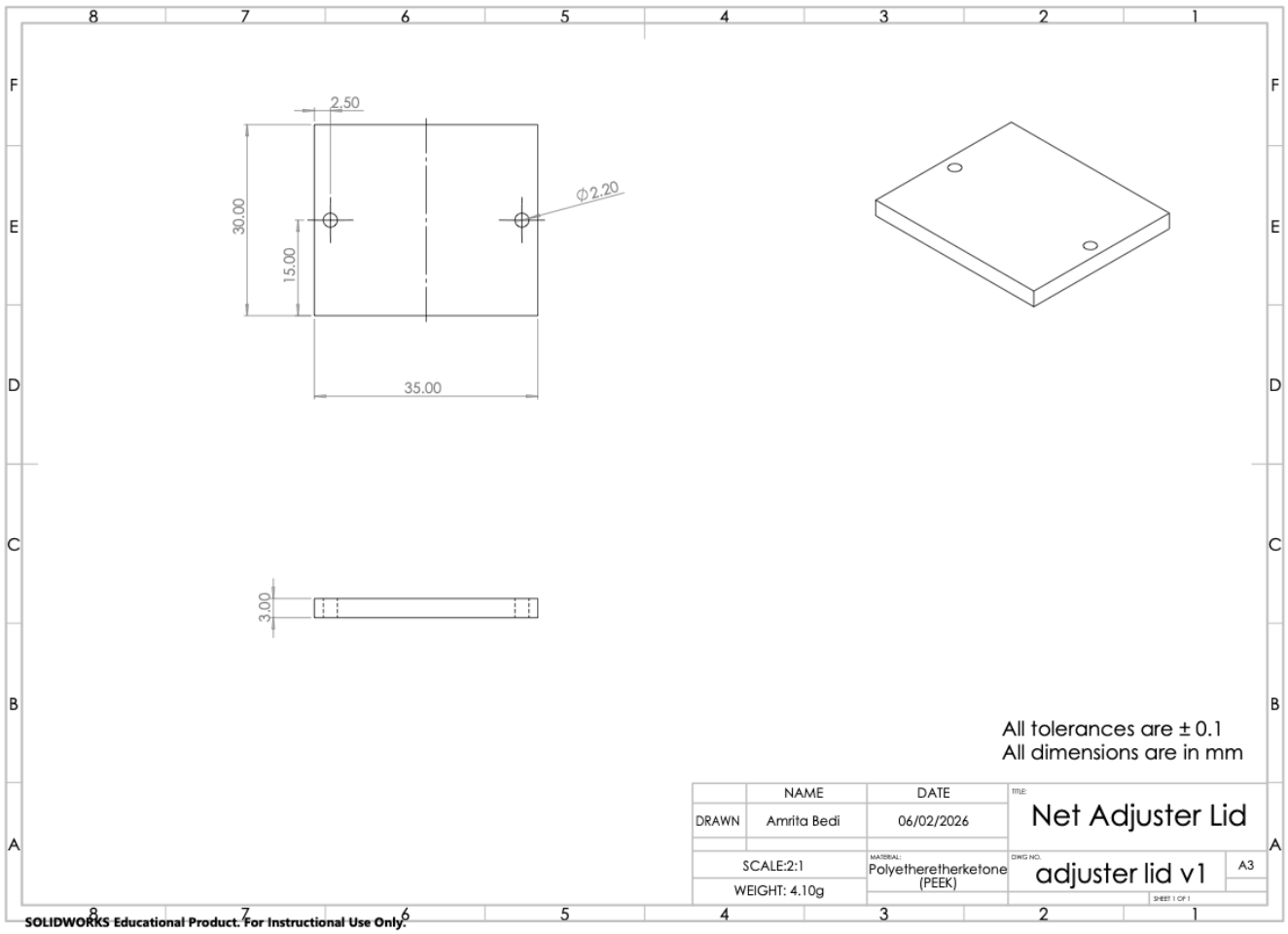
Part Engineering Drawings

Link to tutorial: https://youtu.be/vp34r5lha5o

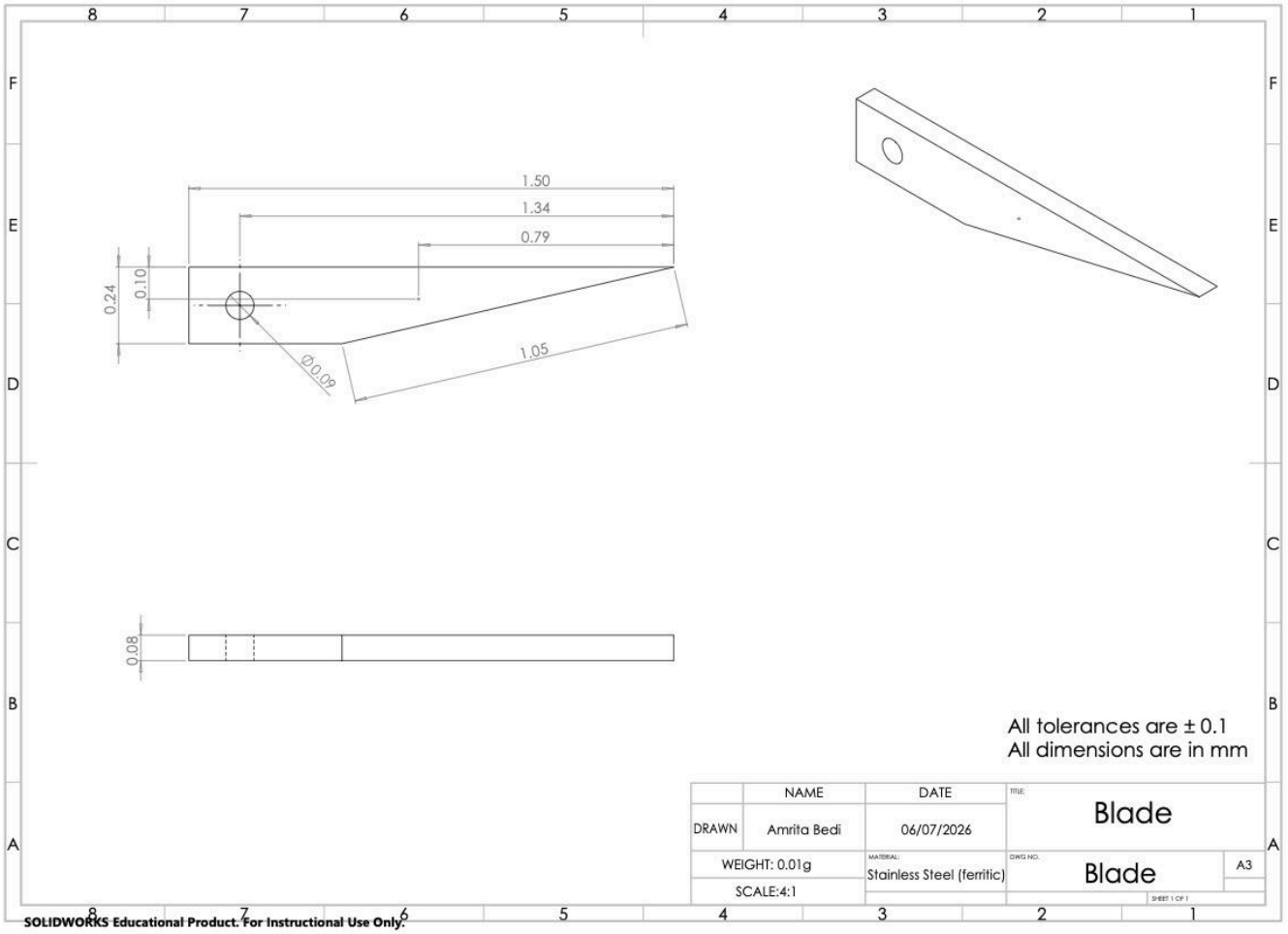
1. Adjuster Base



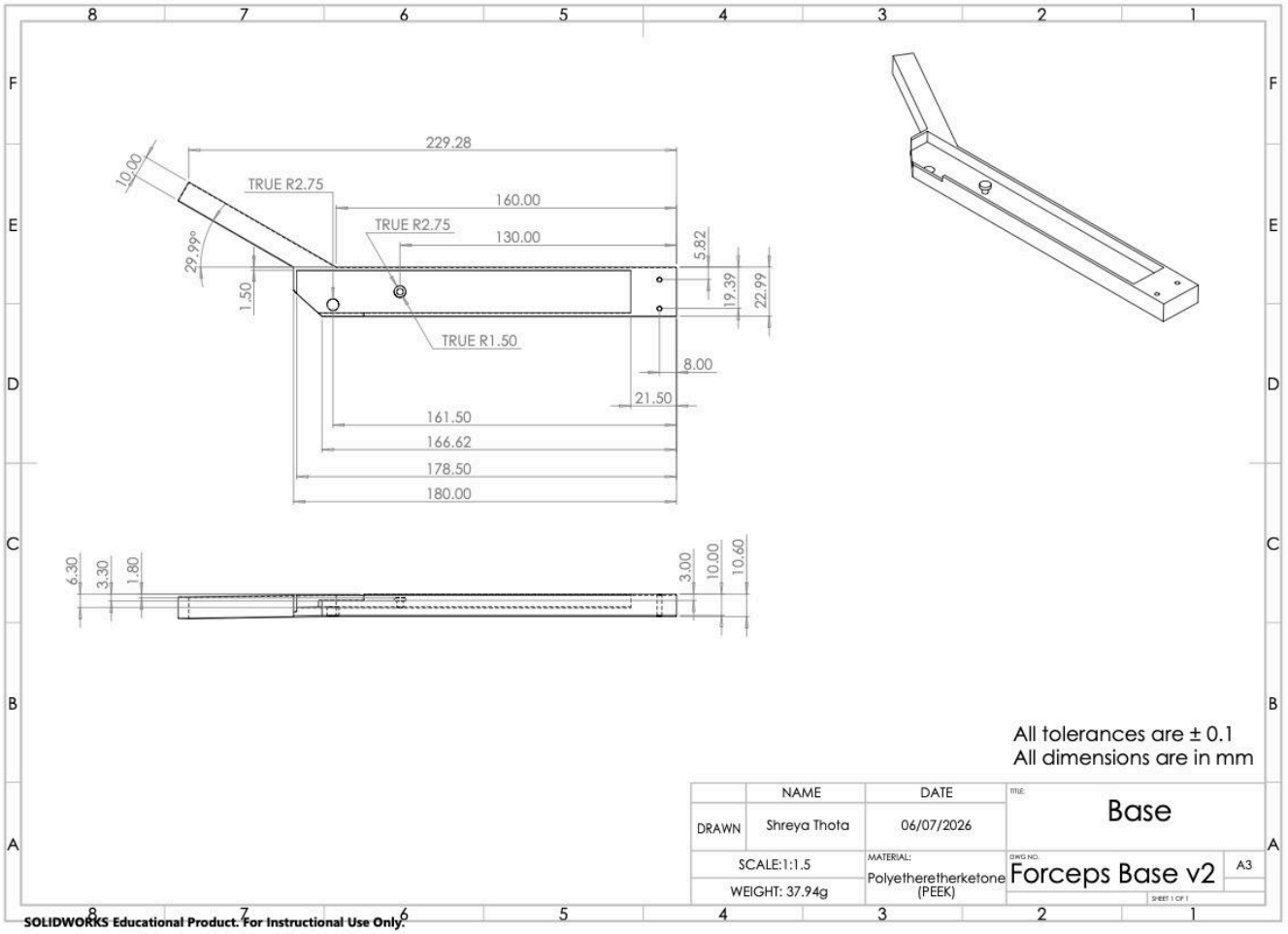
2. Adjuster Lid



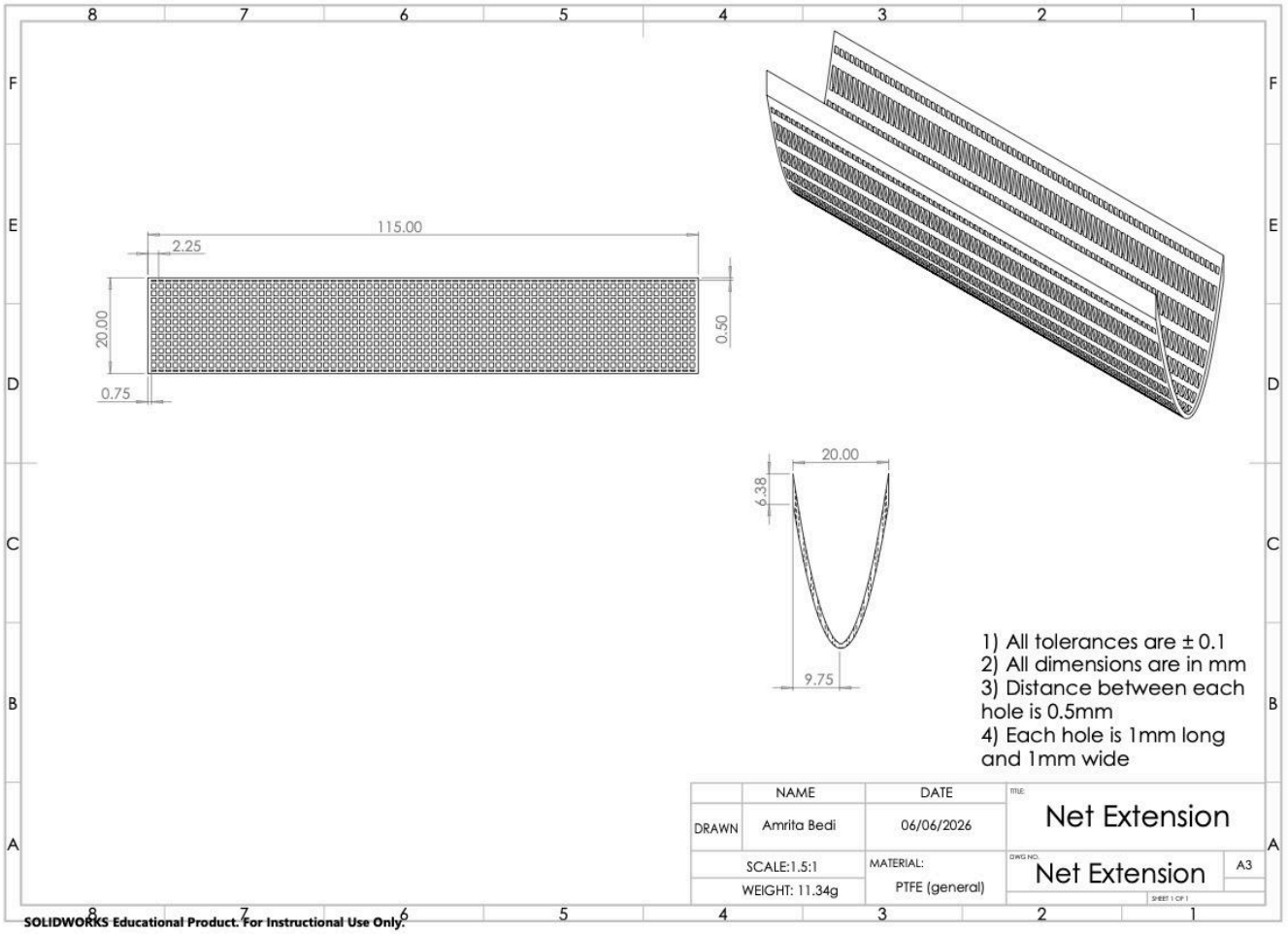
3. Blade


1.50
1.34
0.79
0.24
0.10
Ø0.09
1.05
0.08
All tolerances are ± 0.1
All dimensions are in mm
NAME
DATE
DRAWN
Amrita Bedi
06/07/2026
Blade
WEIGHT: 0.01g
SCALE:4:1
Stainless Steel (ferritic)
Blade
A3
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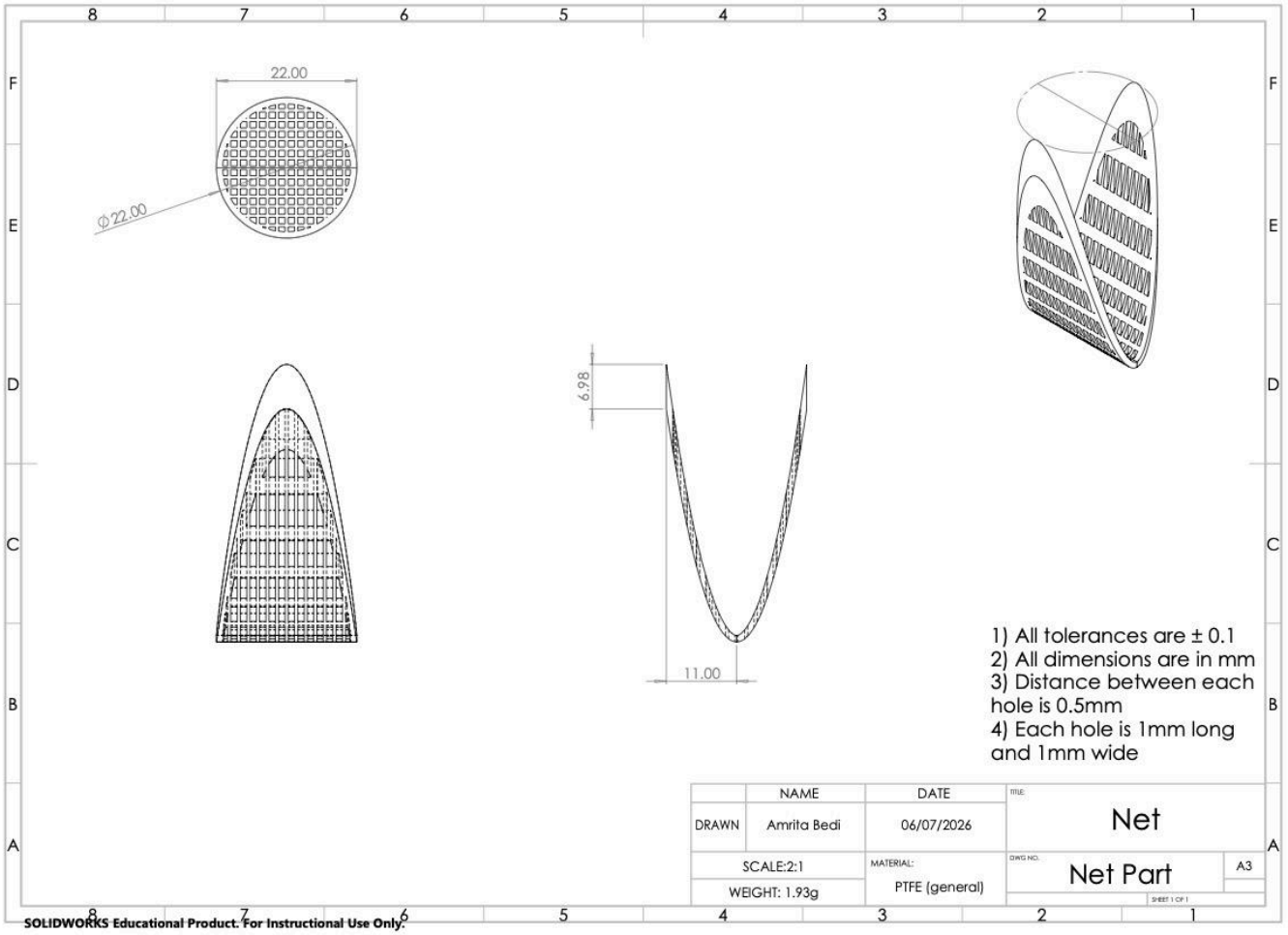
4. Forceps Base



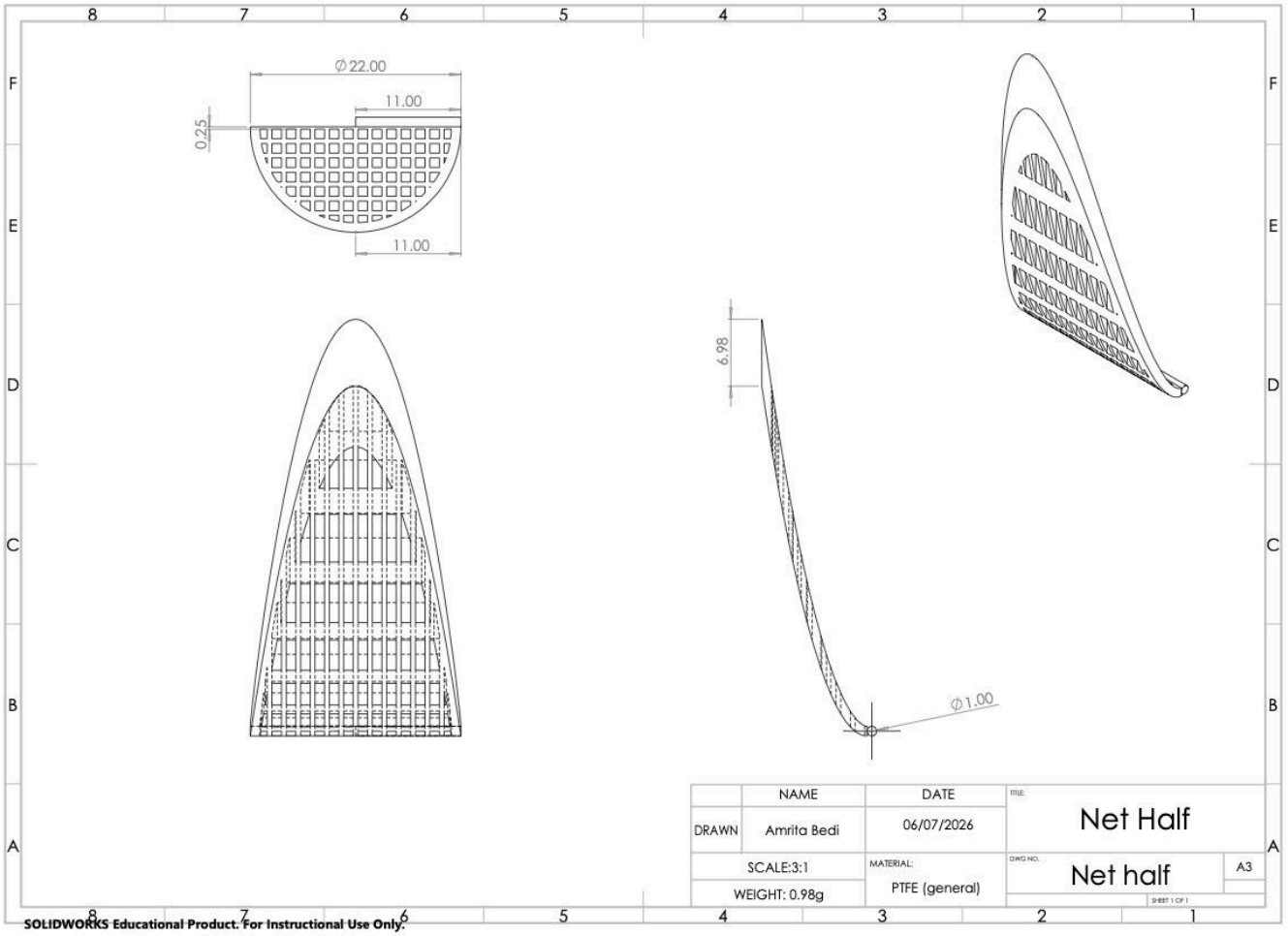
5. Net Extension



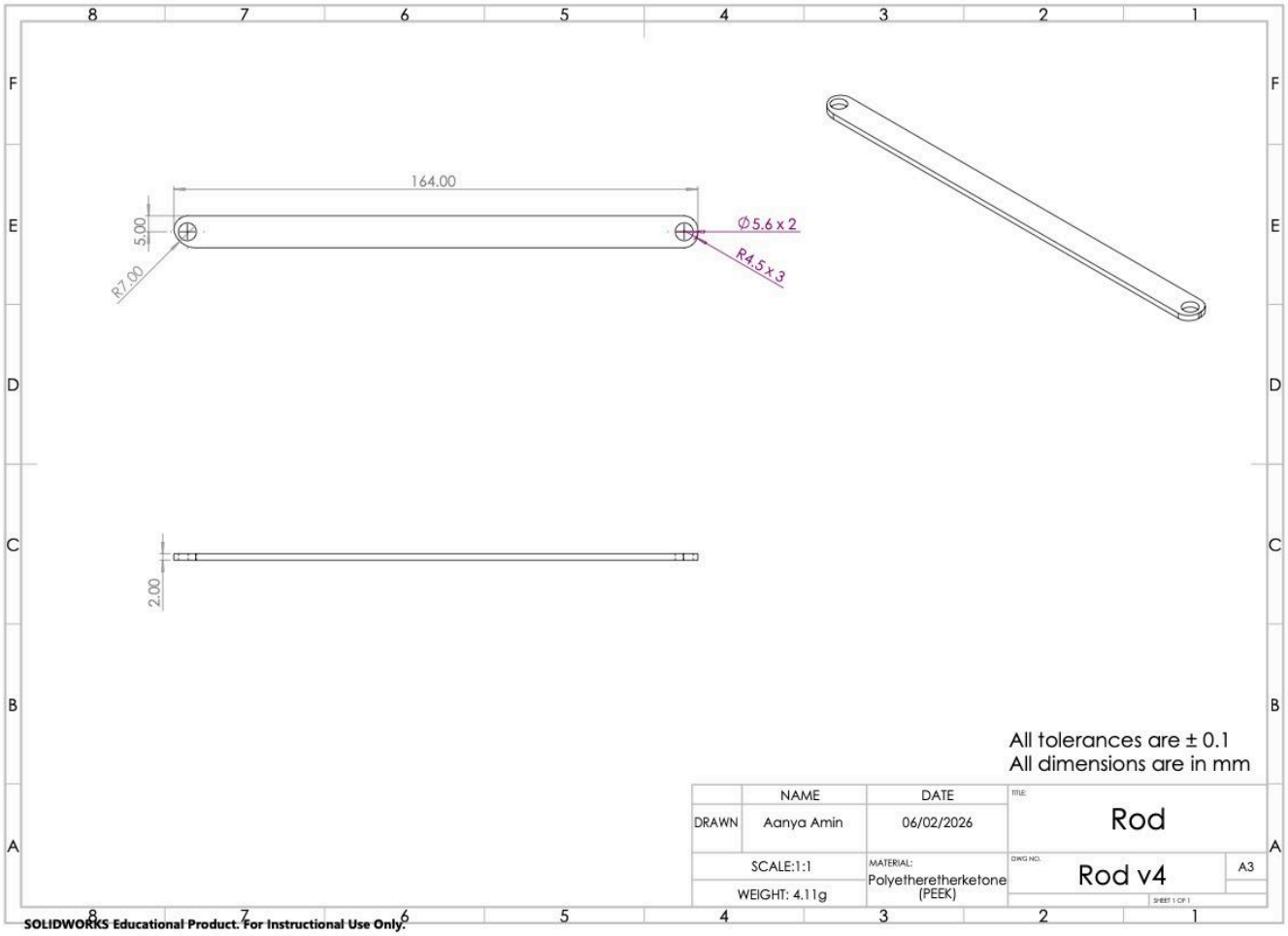
6. Net


22.00
Φ22.00
6.98
11.00
1) All tolerances are ± 0.1
2) All dimensions are in mm
3) Distance between each hole is 0.5mm
4) Each hole is 1mm long and 1mm wide
NAME
DATE
DRAWN
Amrita Bedi
06/07/2026
Net
SCALE:2:1
WEIGHT: 1.93g
MATERIAL:
PTFE (general)
Net Part
A3
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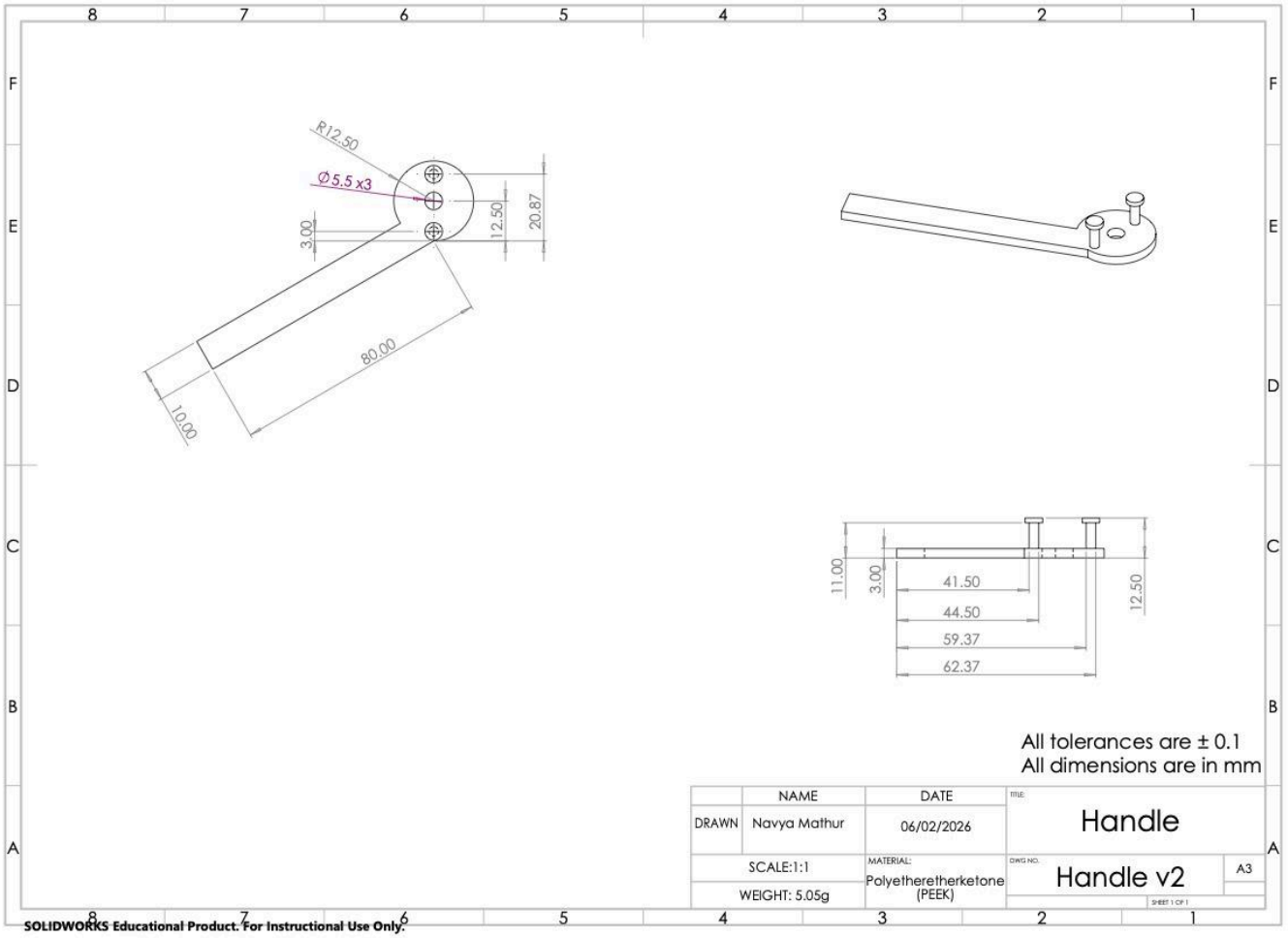
7. Net Half for Hinge Mechanism



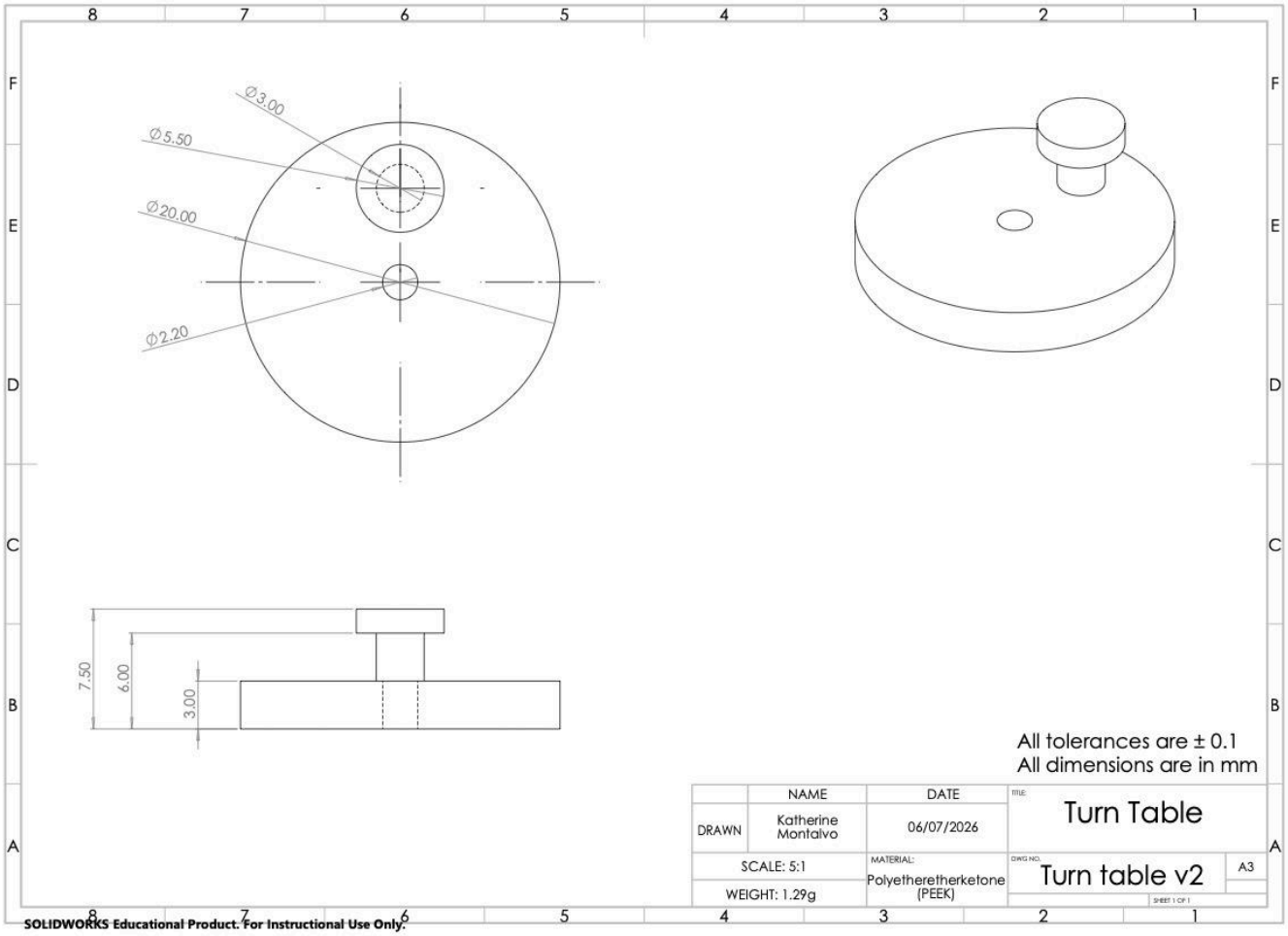
8. Rod


164.00
5.00
R7.00
Ø5.6 x 2
R4.5 x 3
2.00
All tolerances are ± 0.1
All dimensions are in mm
NAME
DATE
DRAWN
Aanya Amin
06/02/2026
TITLE:
Rod
SCALE:1:1
WEIGHT: 4.11g
MATERIAL:
Polyetheretherketone (PEEK)
DWG NO.
Rod v4
A3
SHEET 1 OF 1
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9. Handle

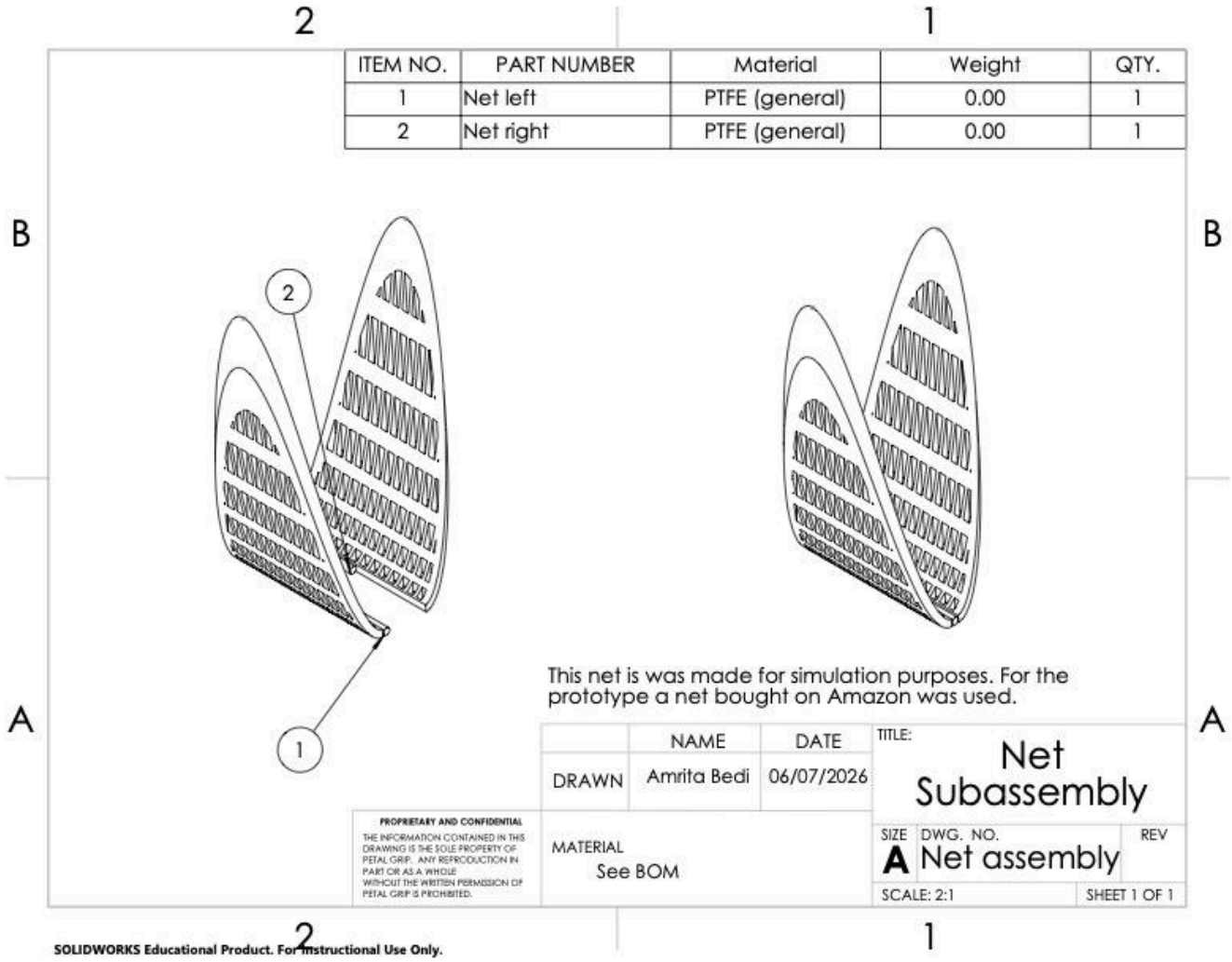

R12.50
Ø5.5 x3
3.00
12.50
20.87
80.00
10.00
11.00
3.00
41.50
44.50
59.37
62.37
12.50
All tolerances are ± 0.1
All dimensions are in mm
NAME
DATE
DRAWN
Navya Mathur
06/02/2026
Handle
SCALE:1:1
WEIGHT: 5.05g
MATERIAL:
Polyetheretherketone (PEEK)
Handle v2
A3
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10. Turntable



Assembly Engineering Drawings

1. Net Subassembly (included in the part creation video)

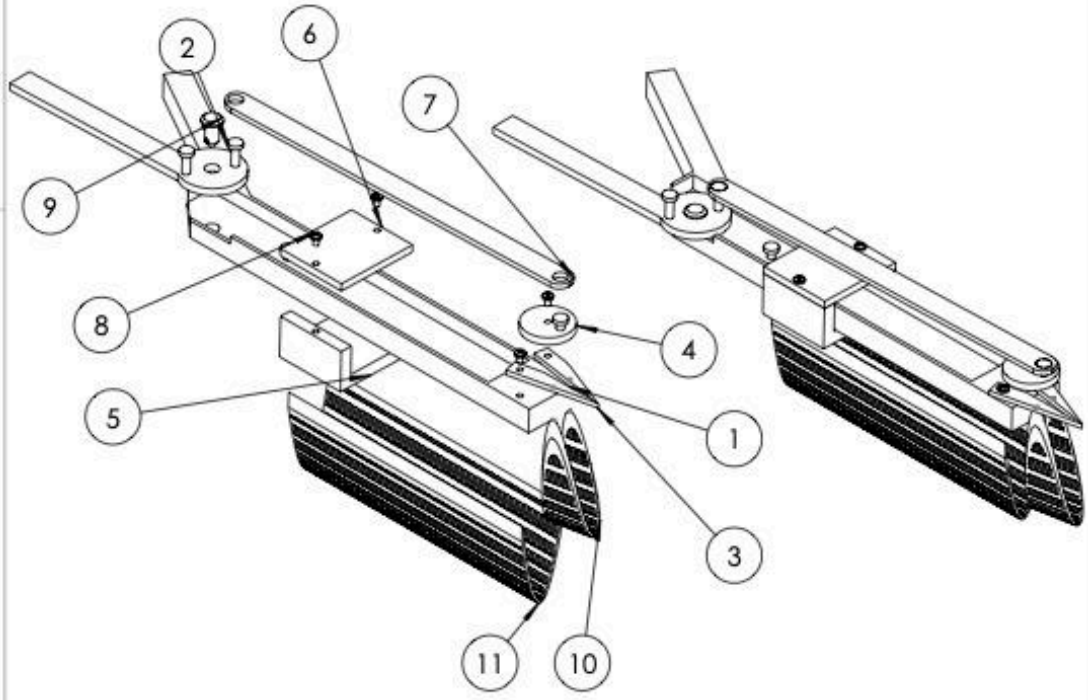


ITEM NO.	PART NUMBER	Material	Weight	QTY.
1	Net left	PTFE (general)	0.00	1
2	Net right	PTFE (general)	0.00	1

2. Forceps Assembly

Link to video: https://youtu.be/iFjUac2jj04

ITEM NO.	PART NUMBER	QTY.	Material	Weight
1	Forceps and Bladeholder v2	1	Polyetheretherketone (PEEK)	37.94
2	Turn Table and Handle v2	1	Polyetheretherketone (PEEK)	5.05
3	Blade	2	Stainless Steel (ferritic)	2.28
4	Turn table v2	1	Polyetheretherketone (PEEK)	1.29
5	adjuster base	1	Polyetheretherketone (PEEK)	11.47
6	adjuster lid	1	Polyetheretherketone (PEEK)	4.10
7	Rod v3	1	Polyetheretherketone (PEEK)	4.11
8	ISO 7045 - M2 x 3 - Z - 3N	4		
9	ISO 2341 - B - 5 x 10 x 1.2 - St	1		
10	Net assembly	1		
11	Net Extension	1	PTFE (general)	11.34



Net, screws, and pivot pins were purchased on Amazon

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	NAME	DATE
DRAWN	Amrita Bedi	06/07/2026

DO NOT SCALE DRAWING

MATERIAL: See BOM

TITLE: Forceps Assembly

SIZE: A | DWG. NO. Forceps Assembly | REV.

SCALE: 1:2 | SHEET 1 OF 1

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Materials and Prototype Designing

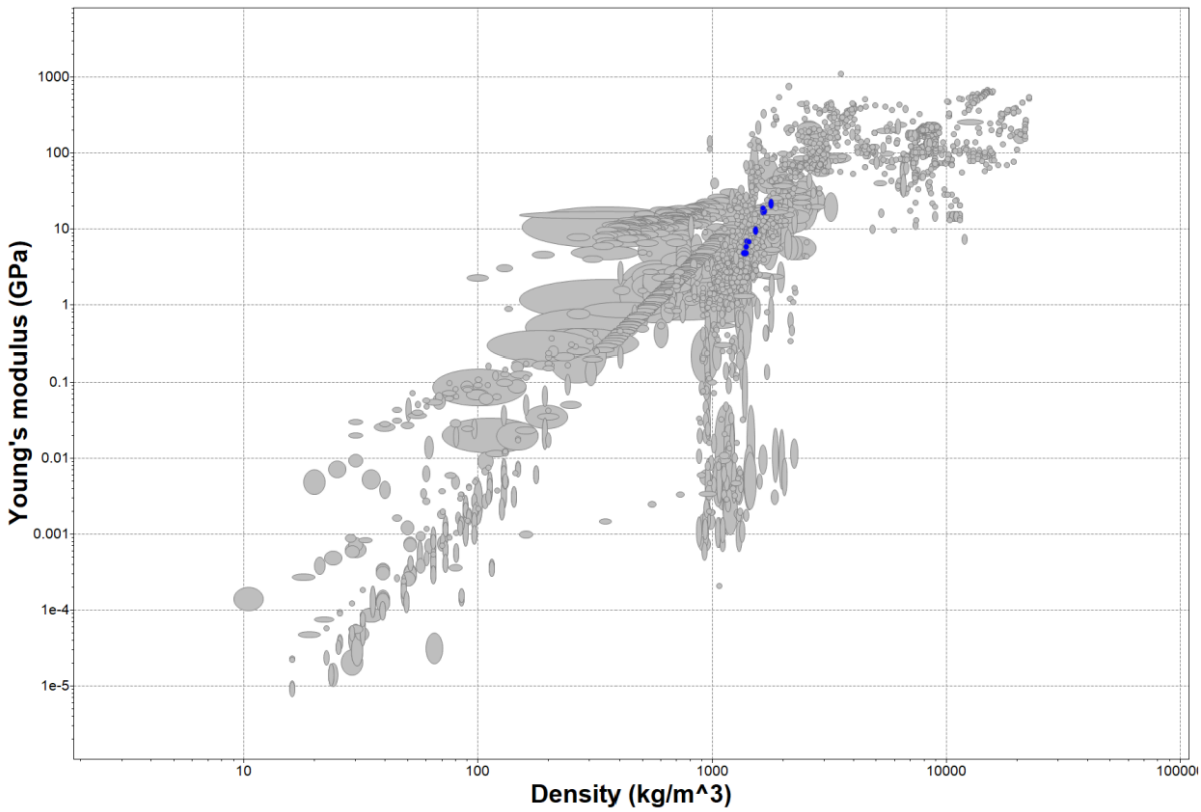
Different materials from level 3: bioengineering on Granta EduPack 2023 were considered as the constraints and requirements for each part of our device were different. All constraints were carefully considered by looking at existing research on medical safety protocols, material properties, and safety standards. **[5, 6, 7]**

Base, Rod, Handle:

The base, rod, and handles were required to provide high stiffness, low weight, durability, and compatibility with medical sterilization. Materials were screened with constraints being density ≤ 5000 kg/m³, Young's Modulus ≥ 2.9 GPa, fracture toughness ≥ 7 kJ/m², USP Class VI / ISO 10993 compliant, maximum service temperature of at least 134°C ,and good/excellent steam autoclave sterilizability. **[8]**

An Ashby chart of Young's Modulus versus Density was used to identify lightweight materials with high stiffness. Metals were excluded despite their high stiffness because their high thermal conductivity can make them feel cold during patient contact, reducing comfort. Polymers also offer lower weight and greater design flexibility.

Stage 1:	Density (kg/m^3), Young's modulus (GPa)	Summary

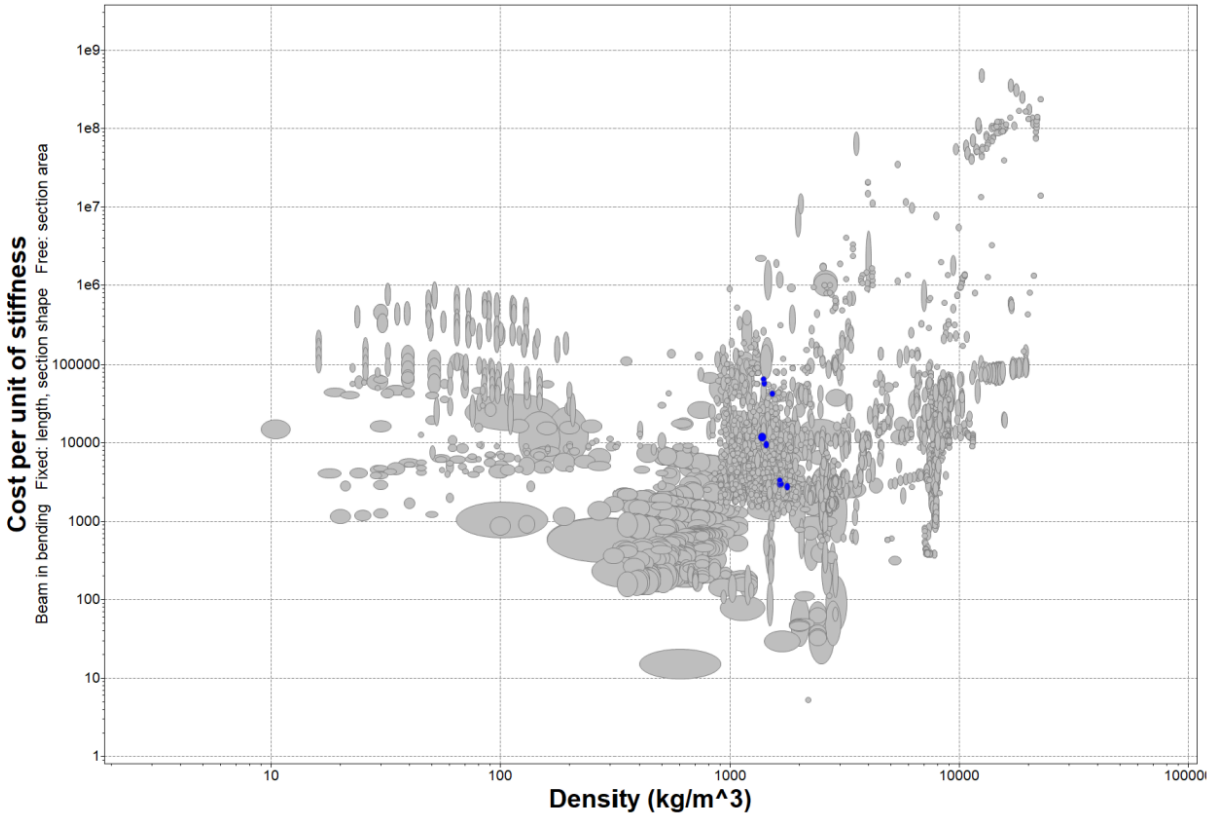


The housing was then modeled as a beam in bending, as the primary requirement is resistance to bending deformation. The objective was to minimize cost per unit stiffness, producing the performance index:

$$M = \frac{C_m \rho}{E^{1/2}}$$

where C_m is material cost, ρ is density, and E is Young's modulus. Materials with lower values provide greater stiffness at lower cost.

Stage 3:	Density (kg/m^3), Cost per unit of stiffness	Summary

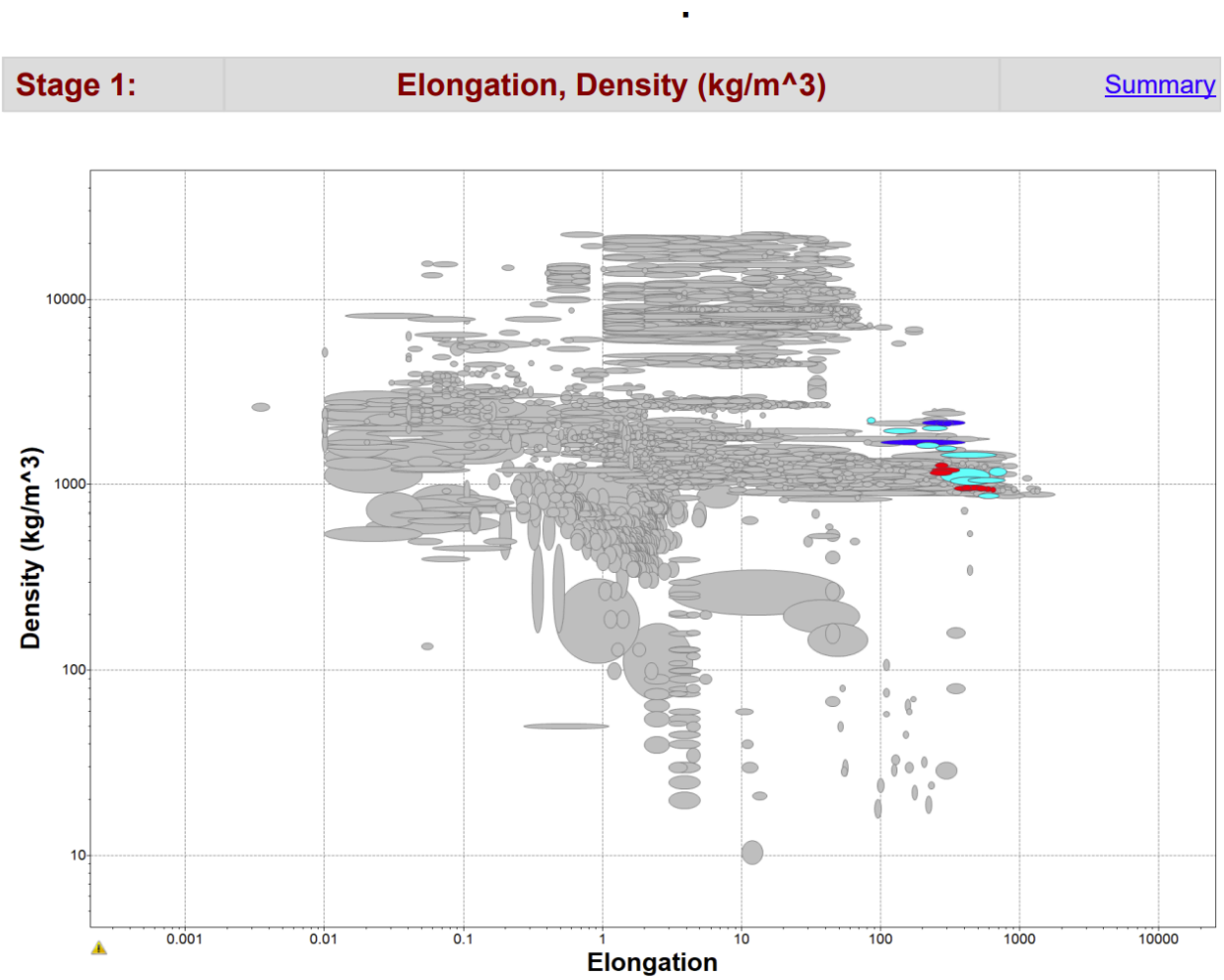


Although several glass-fiber-reinforced polymers ranked highly, PEEK reinforced with 30% glass fiber was selected. PEEK offers excellent stiffness, toughness, dimensional stability, and resistance to repeated steam-autoclave sterilization. It is also widely used in medical devices due to its biocompatibility and long-term durability. Therefore, PEEK (30% glass fiber) provided the best balance of structural performance, sterilization resistance, patient comfort, and medical-device suitability for the housing. **[10]**

Mesh:

A maximum Young's modulus of 1 GPa was selected to ensure the net remains sufficiently flexible during insertion and retrieval. Materials with lower modulus values undergo greater elastic deformation under an applied load, whereas materials with higher modulus values are more rigid and resist stretching. For a deployable and adjustable medical net, excessive stiffness

could hinder expansion and increase the risk of tissue damage. Density was limited to 3000 kg/m³ to minimise bulkiness. A maximum service temperature of at least 134°C was required to withstand standard steam-autoclave sterilization cycles used for reusable medical devices. Finally, materials were required to satisfy ISO 10993 biocompatibility requirements and be available in USP Class VI medical grades to ensure safe patient contact. The Ashby chart was used for all these constraints and to select the best material considering minimum density and maximum elongation. **[9]**

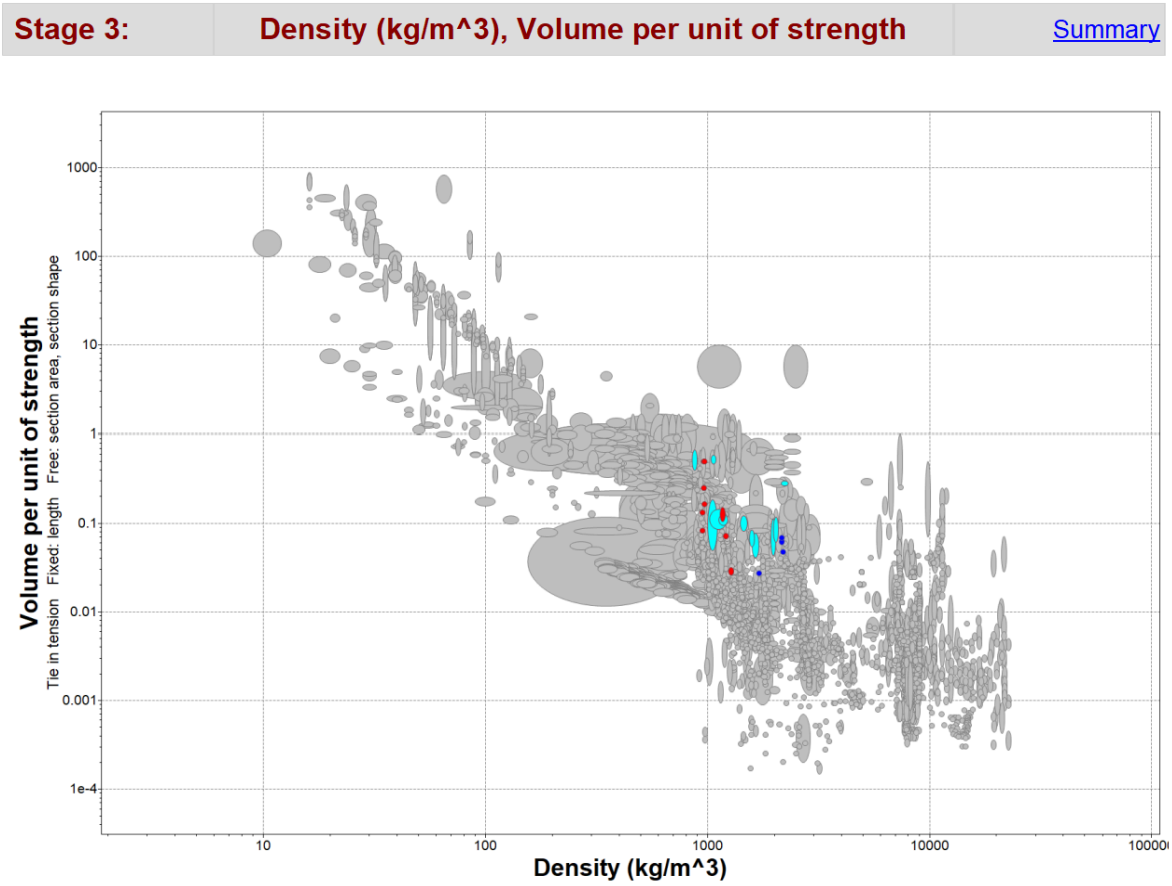


Then, the net was modeled as a tie in tension because the mesh strands primarily experience tensile forces during deployment, retrieval, and tissue capture. The function and loading condition "Tie in tension" was selected with length fixed and section area/shape free to vary. The

limiting constraint was set to strength, while the objective was to minimize volume. This generated the performance index:

$$M = \frac{1}{\sigma y}$$

The resulting axis, Volume per Unit Strength, ranks materials according to the minimum material volume required to carry a given tensile load. Materials with lower values provide greater strength efficiency and are therefore preferred for lightweight net structures.



The resulting candidates included ETFE, TPC, fluorinated elastomers, and PTFE. Although ETFE exhibited the highest performance index ranking, PTFE was selected as it passed all screening constraints for the net, including density ≤ 3000 kg/m³, Young's modulus ≤ 1 GPa, maximum service temperature ≥ 134°C, medical-grade availability, and good/excellent steam autoclave sterilizability. In the selection results, PTFE showed a high elongation range of 200-400%, meaning it can undergo large deformation during net deployment without brittle failure.

Cost Analysis and Results

Quantity	Item	Cost
1 Pack	Nuts and Bolts **Diameters:** 3 mm, 4 mm, 5 mm, 6 mm **Lengths**: 7.6 mm, 11.171 mm, 15.44 mm 19.41 mm	$10
1 Pack	Corby Bolts, Pivot Pins **Diameter**: 5 mm **Length**: 9 mm	$10.80
1 Pack	Black Carbon Steel Sharp Point Screw **Diameters:** 2 mm, 3 mm, 4 mm	$9.49
1 Pack	Small Compression Spring **Diameter**: 3 mm **Length**: 10 mm	$13.30
1 Pack	Small Compression Spring **Diameter**: 4 mm **Length**: 10 mm	$13.30
1	26 Gauge Wire **Diameter**: 0.4 mm **Length**: 33 ft	$5.39
1	Ultra Fine Screen Mesh **Length:** 10 ft **Width:** 4 ft	$6.99
1	3D Printer Super Glue **Volume**: 1.75 fl oz	$12.99
1	Craft Knife, Extra Blades	$4.99

	Angle: 25°	
1	3D-Printing (PLA) **Printing Iterations**: 6 **Total Print time**: 3.4 hrs	$20.40
Total:		**$107.65**

The prototype assembly was constructed using 3D-printed PLA components and commercially available hardware. The stationary blade was attached directly to the base and the rotating blade was mounted to a turntable mechanism both secured using 2 mm screws. The adjustable handle with a turntable was connected through a 5mm pivot pin and spring system, enabling smooth rotation of the device. The upper and lower turntables were integrated using a 3D-printed rod with pin features and holes to secure the assembly. The mesh capture mechanism was attached using adhesive and connected to the retracting slide mechanism, which was mounted directly to the base.

The total purchasing cost of commercially available materials was $87.25, including screws, springs, pivot pins, glue, wire, and mesh fabric. The forceps underwent six prototype iterations, resulting in a total print time of 3.4 hours at a printing cost of $6 an hour for a total of $20.40. These iterations allowed for improved fit between components and increased functionality of the design. This brings the total cost of our design to $107.65, remaining below the $215 allocated budget while establishing a functional and cost-effective design.

1. Methodology

A static stress analysis was performed using SolidWorks Simulation to evaluate the structural integrity of the redesigned cervical biopsy forceps under expected operating conditions. When using this device, the physician applies a force to the rotating handle, causing the turntables to rotate and thereby move the linkage rod. The linkage rod then transfers the force to the biopsy jaws, resulting in blade closure and snipping the tissue to be collected.

A force of 40 N was applied to the moving handle to simulate the force exerted by a clinician during cervical biopsy collection. The selected force was based on values reported in the literature for hand-actuated surgical instruments, including laparoscopic scissors and forceps, which experience comparable squeezing forces during operation. Therefore, 40 N was chosen as a realistic clinical loading condition for evaluating the structural integrity of the device under expected use. **[11]**

The methodology for the studies we completed is as follows:

- **Static Study:**
    - Simplified Geometry: Removed non-critical geometry to focus on specific interactions between key parts.
    - Material Applied: PEEK was applied to the base, rod, and handles, while stainless steel was used for all other components, like the blades.
    - Fixture Used: Fixed certain faces of our parts to simulate what would occur during the physician's usage of these forceps.
    - Loading of Forces: Applied a force to the lower rotating handle’s right face, replicating how the handle is pressed.
    - Mesh: Fine mesh near critical points of contact to capture stress concentration accurately.
    - Study Type: Mixed static studies for load-holding evaluation and motion studies for release dynamics.
    - Acceptance Criteria: Compare stresses on individual parts and multi-part systems against the yield stress of the part material.

- **Motion Study:**
    - Mates & Constraints: Applied motion-driving and limit mates to control the movement ranges of key parts based on real-world behavior.
    - Force Inputs: Modeled expected forces/torques using linear motors with expected magnitudes.
    - Gravity & Contacts: Enabled gravity and relevant contact sets to model interactions between parts during motion.
    - Timeline Setup: Defined motion segments using keyframes to simulate accurate timing of user input.
    - Acceptance Criteria Evaluation: Compare motion dynamics of the handle to ensure it is moving the linkage rod, which then moves the blades

Previous design iterations had significant bending of the linkage rod, indicating that it was the most critical load-bearing component and a potential point of failure. Because the linkage rod carries most of the tensile load, a simulation study was conducted to determine whether the redesigned device could safely withstand clinical operating forces.

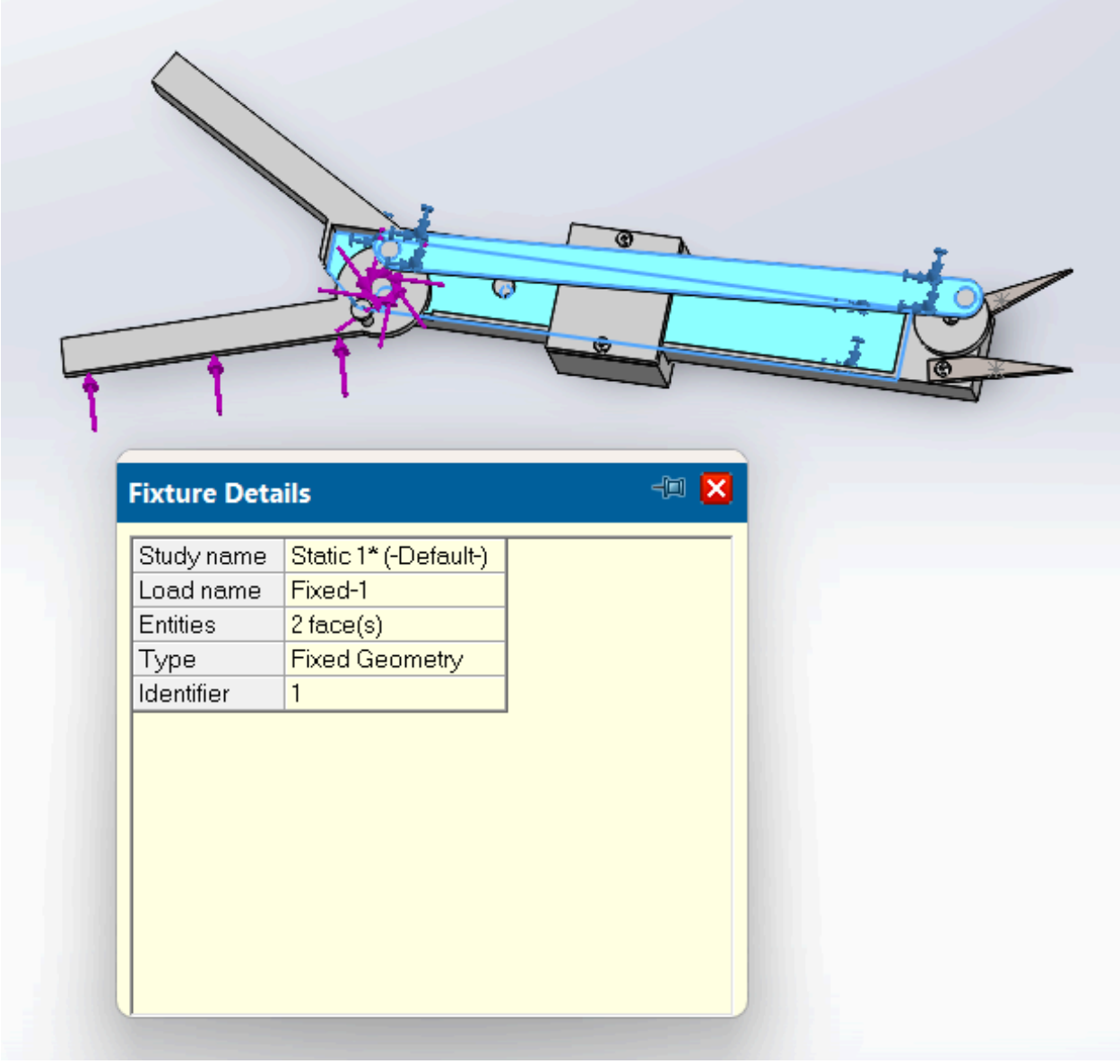
Link to motion study video - https://youtu.be/lUmuVqZAFyg

2. Why Simulation Study

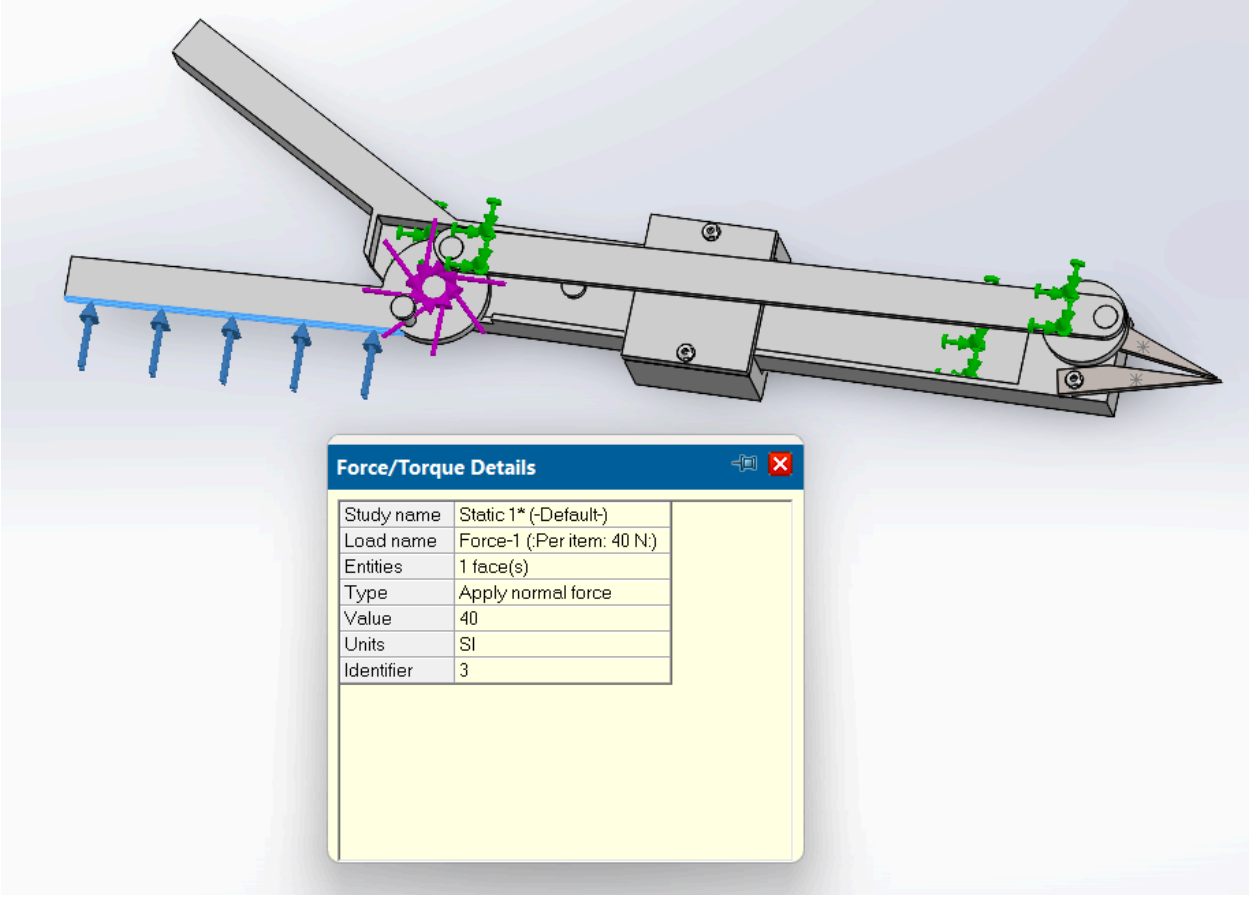
The simulation was necessary to determine whether our cervical biopsy forceps could safely bear the forces during sampling. There is tension being transferred from the handle to the linkage rod, which previously bent(close to snapping) in previous iterations, as it wasn't able to handle the force. The simulation study further confirmed the rod won't snap during the procedure and that all the forces are balanced throughout the device. Maximum von Mises stress must remain below the material yield strength. Maximum displacement must not interfere with jaw closure or tissue capture.

3. Results

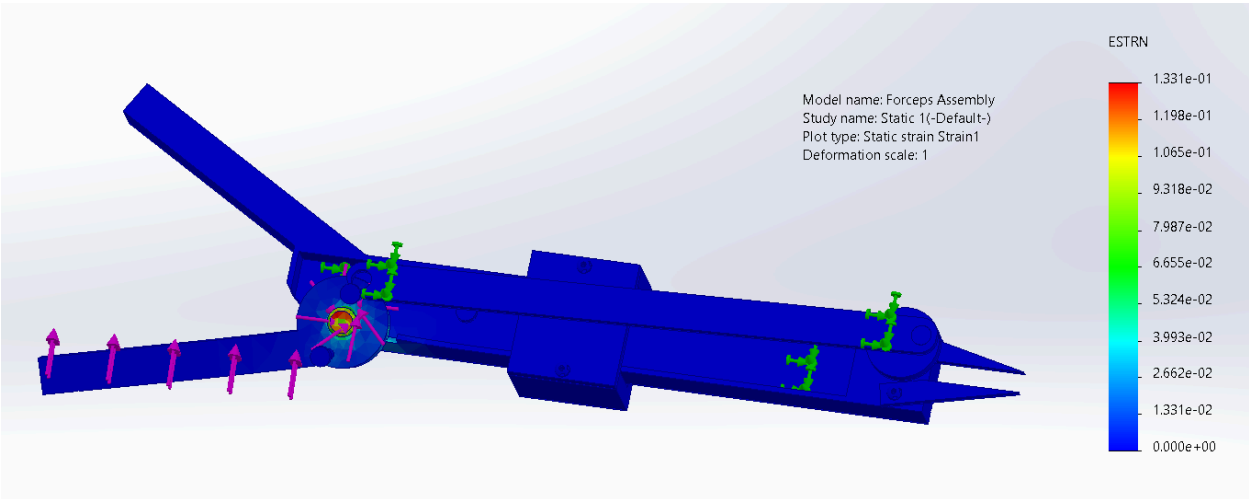
Fixed Geometry:



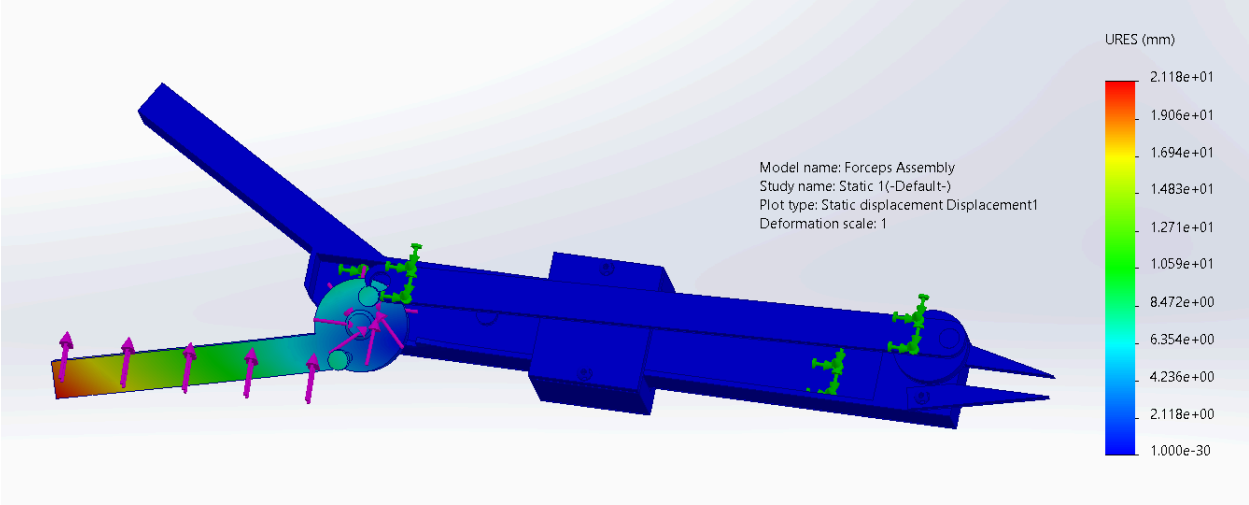
Forces and Torque:



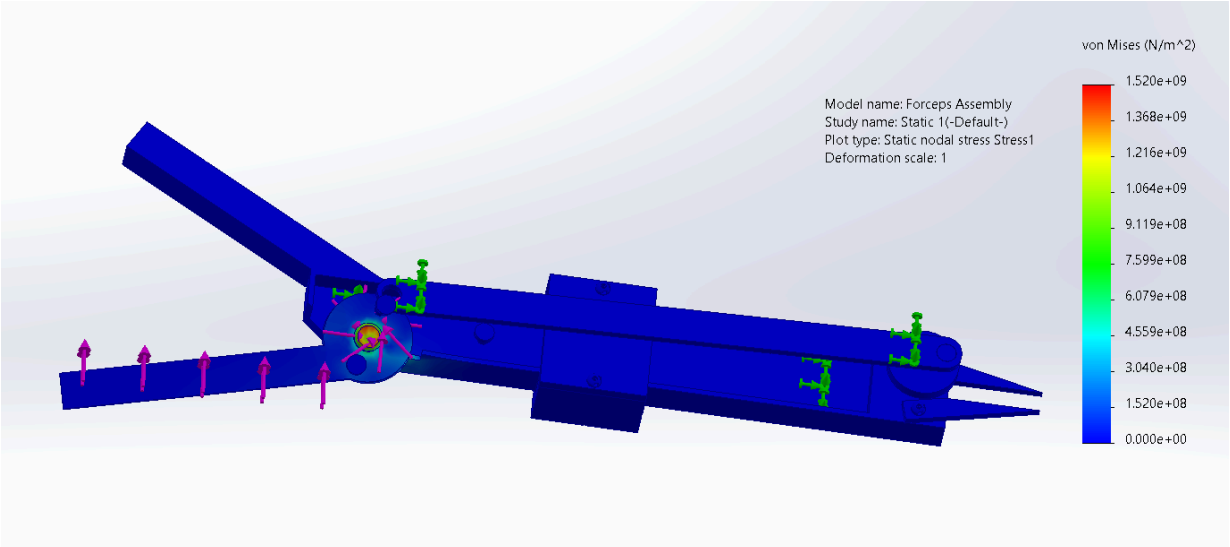
Static Strain Simulation Test



Static Displacement Test:



Static Stress Test:



Individual Analysis

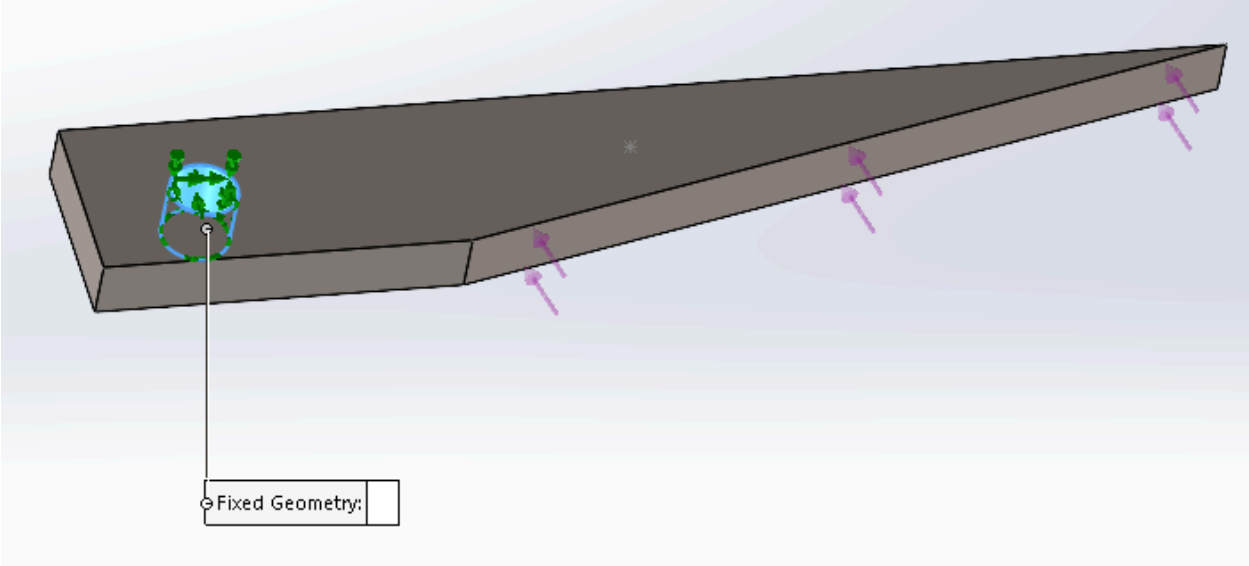
Blade

Simulation of the blade is necessary because it serves as the primary component of the device and allows the user to cut the tissue sample. The blade is connected to the moving turntable and experiences force at the cutting edge when it contacts the tissue. For the simulation, the blade was fixed at the mounting hole to represent the screw or pivot connection that constrains the blade during operation. A force of approximately **14 N** was applied at the cutting edge to represent the estimated tissue resistance force during sampling. This value was selected based on published studies on surgical blade cutting of soft biological tissues, which show that cutting force depends on tissue type, blade geometry, blade sharpness, cutting angle, and insertion/cutting conditions.

Since cervical tissue is a soft biological tissue and the blade is intended to remove a small biopsy sample, 14 N was chosen as a reasonable loading condition to approximate the resistance experienced by the blade during cutting **[12]**. The objective of the simulation is to determine whether the blade can withstand cutting forces without significant deformation near the mounting hole or cutting edge. The simulation results indicate that the stresses remain below the material limits and that displacement is minimal. Therefore, the blade can tolerate the expected tissue-cutting forces without structural failure.

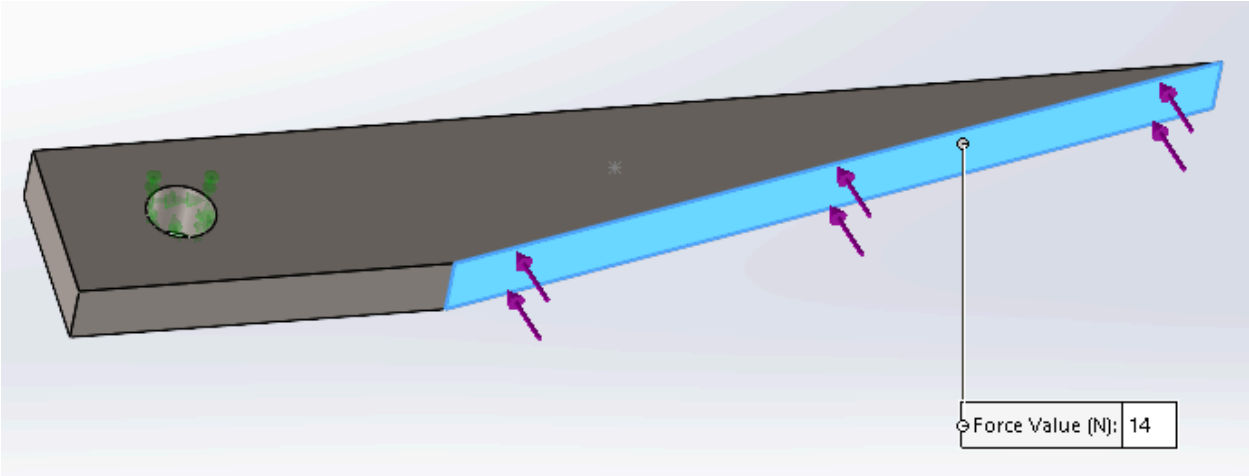
Material: PEEK (30% glass fiber)

Fixture: fixed geometry

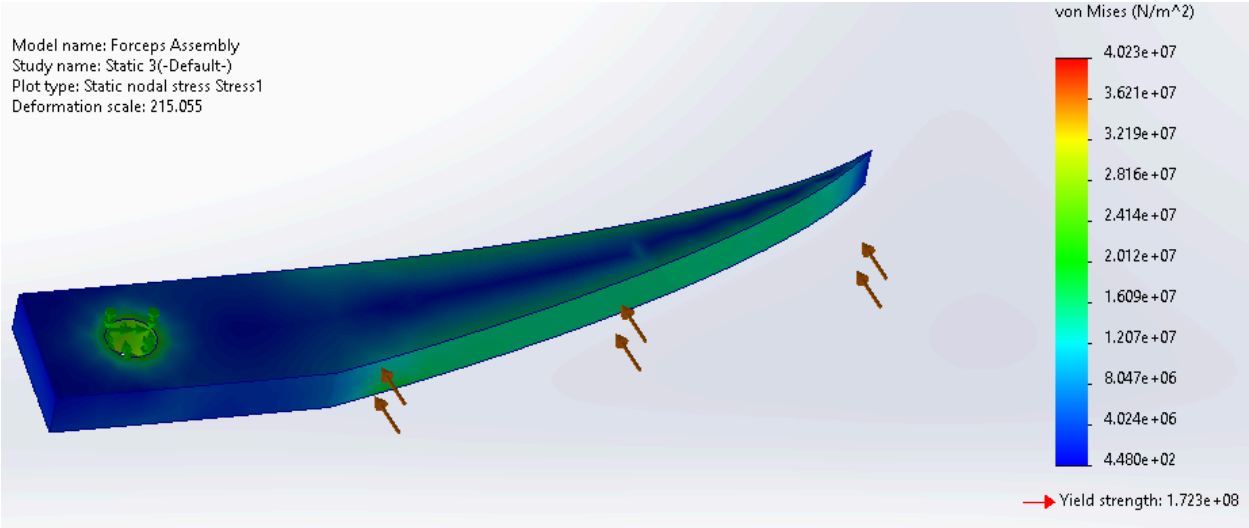


Force: 14 N

Force direction:



Stress plot with deformation:



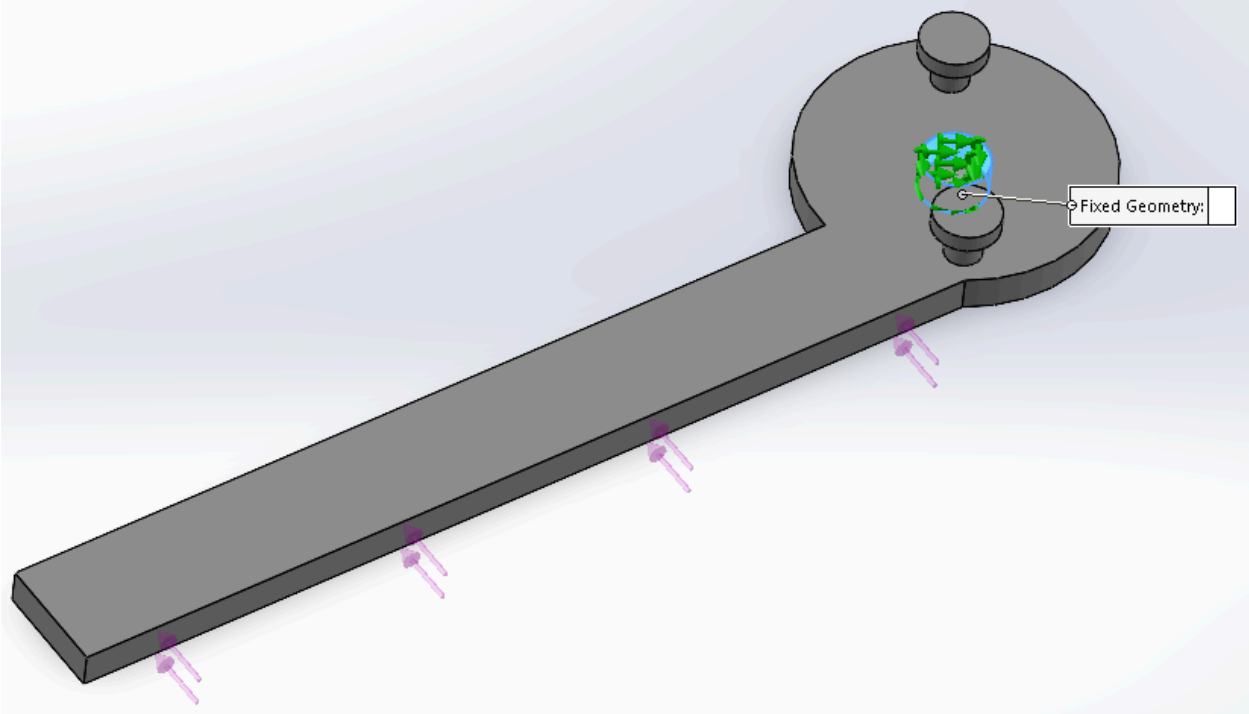
Turntable Handle

Simulation of the turntable handle is necessary because it converts user input force into rotational motion that drives the linkage system. The handle is connected to the base through a pivot pin located at the center of the turntable. During use, the clinician applies a force of approximately **40 N** at the end of the handle. Since this force is applied at a distance from the pivot, it generates a moment about the pivot and causes the handle to rotate. For the simulation, the center pivot hole was fixed to represent the pin connection to the base, while the 40 N force was distributed along the handle grip region to represent user input during device actuation.

The applied force was selected using data from published studies on manually operated surgical instruments, such as laparoscopic forceps and scissors, which are subjected to similar hand-squeezing loads during use. As a result, a load of 40 N was considered a reasonable representation of clinical operating conditions for assessing the strength and performance of the turntable handle **[11]**. The purpose of this analysis is to verify that the handle can withstand repeated loading without excessive deformation or failure near the pivot location. The simulation results indicate that the stresses remain within acceptable limits and that displacement is primarily located at the free end of the handle, which is expected for a lever-like component under normal operation.

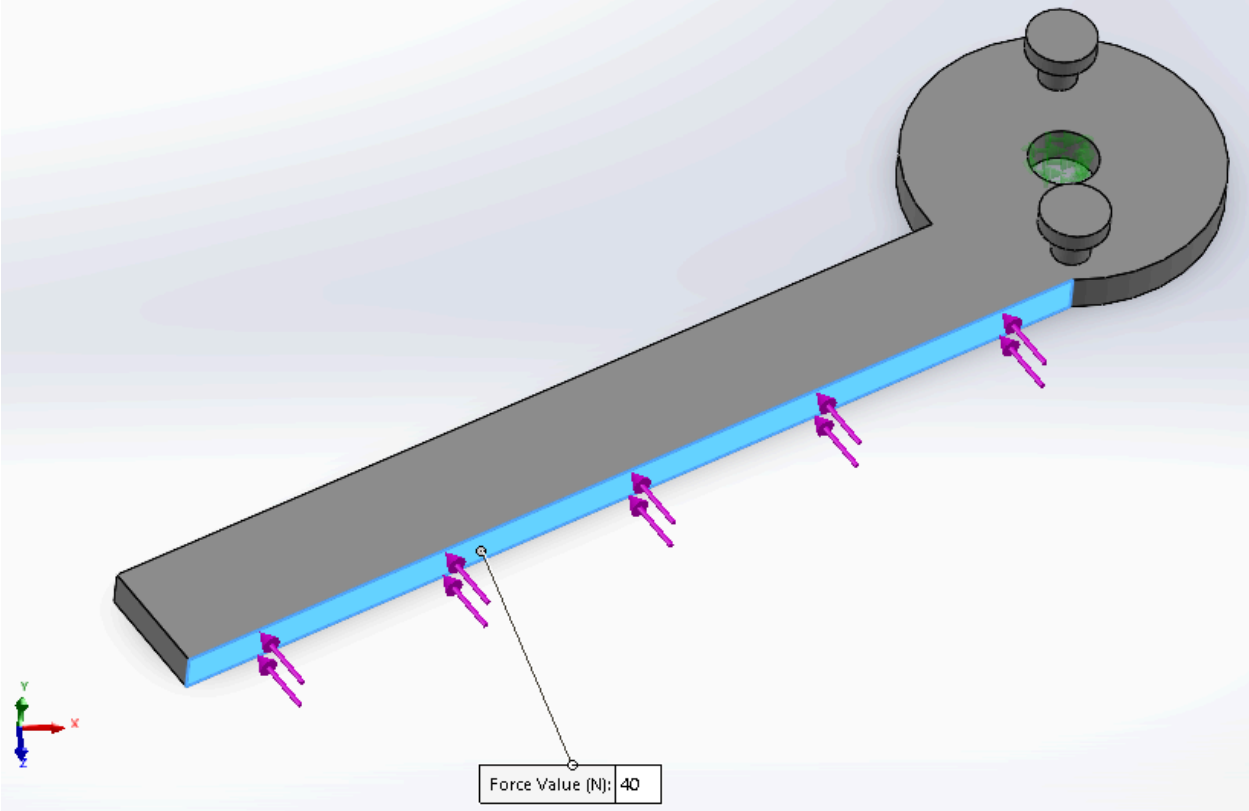
Material: PEEK (30% glass fiber)

Fixture: fixed geometry

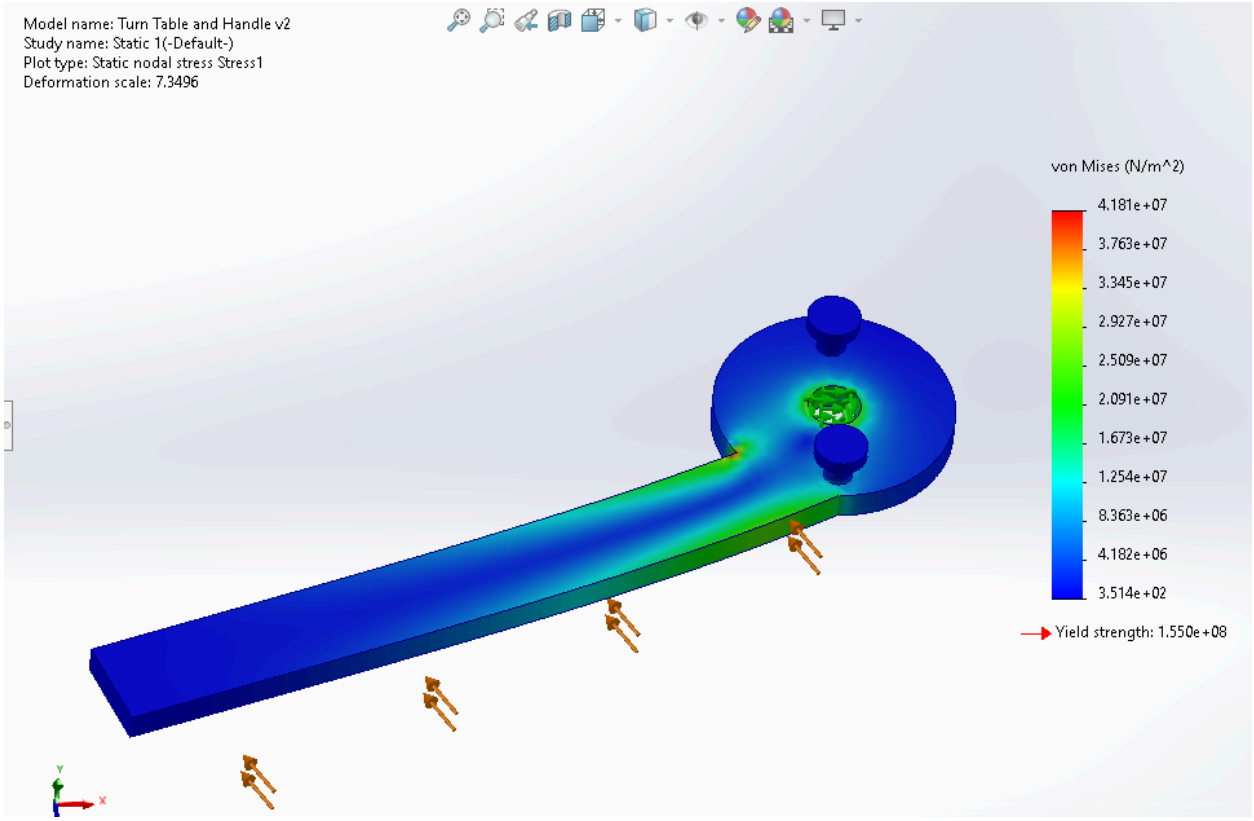


Force: 40N

Force direction:



Stress plot with deformation:

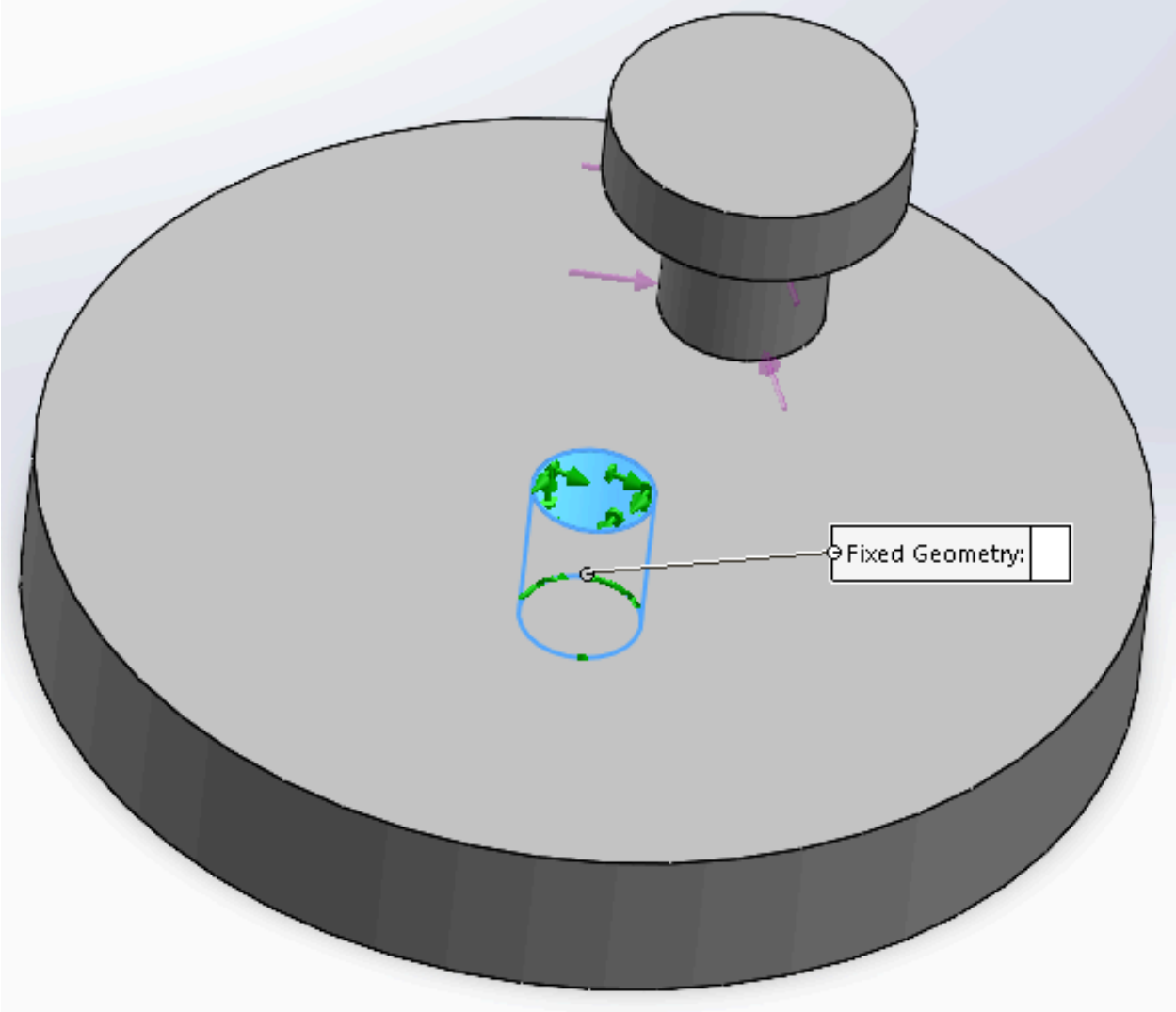


Turntable

Simulation of the upper turntable is necessary because it serves as the connection point between the linkage rod and the moving blade assembly. Motion from the handle is transferred through the linkage rod to the turntable, causing it to rotate and drive the cutting mechanism. For the simulation, the center pivot hole was fixed to represent the screw or pin connection to the base, while a force of approximately **40 N** was applied at the hook where the rod attaches to the turntable. This force represents the transferred load from the 40 N handle input during device actuation. The objective of the simulation is to determine whether the turntable can withstand the transmitted forces without significant deformation around the pivot or rod attachment locations. The results indicate that the stresses remain below the material limits and that displacement is primarily located near the outer regions of the turntable, which is expected during loading. Hence, the turntable remains within acceptable limits and is able to support the expected transferred force.

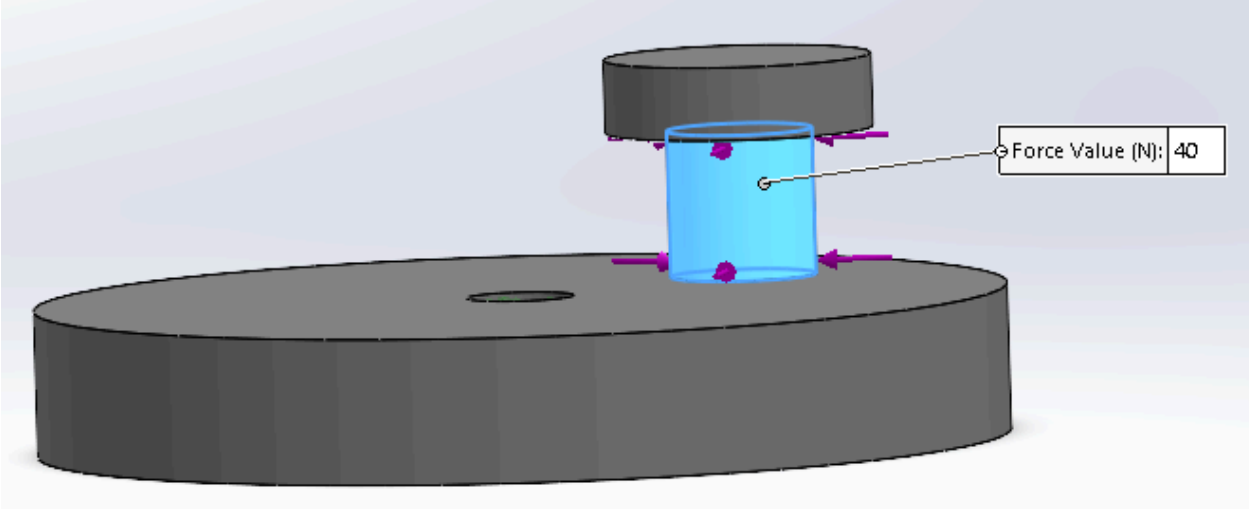
Material: PEEK (30% glass fiber)

Fixture: fixed geometry

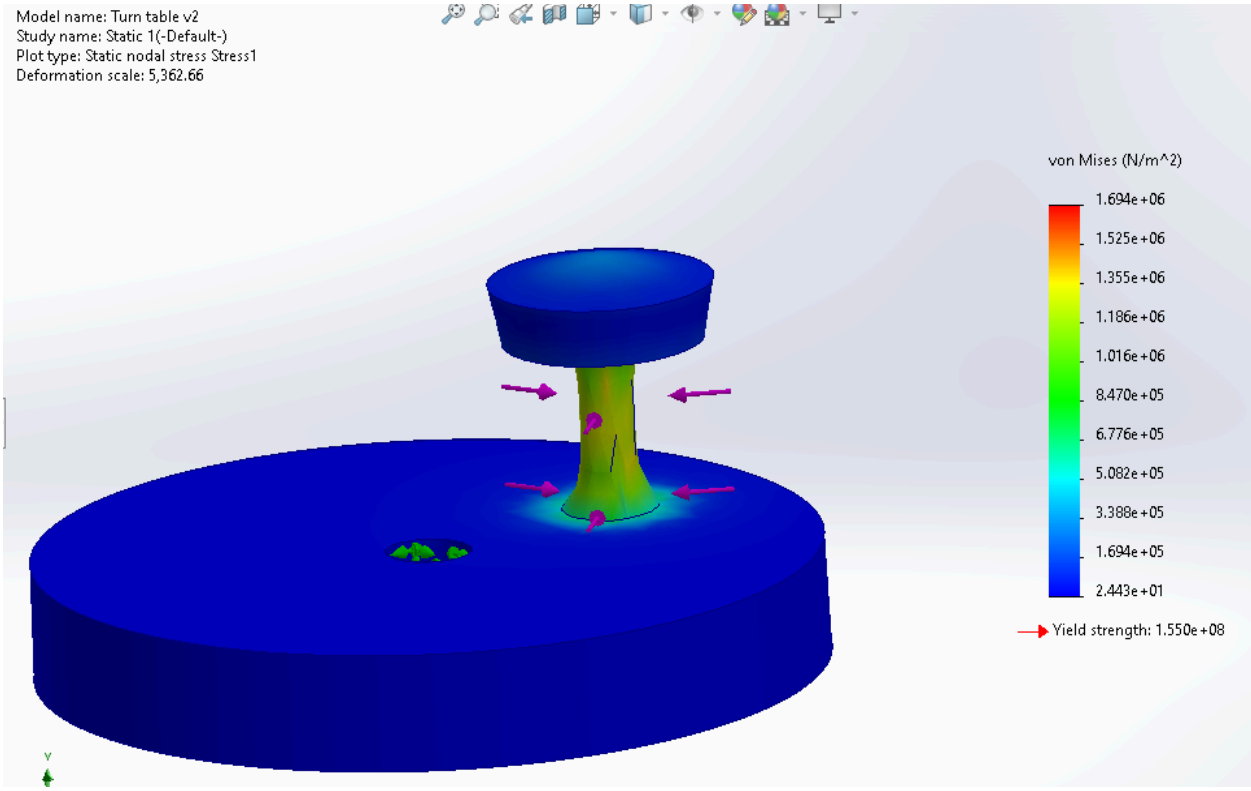


Force: 40 N (transferred load from 40N handle input)

Force direction:



Stress plot with deformation:

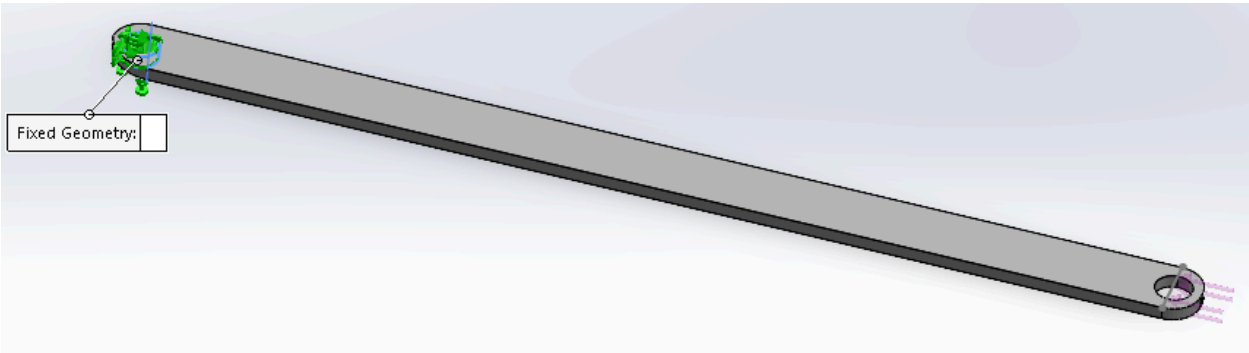


Rod

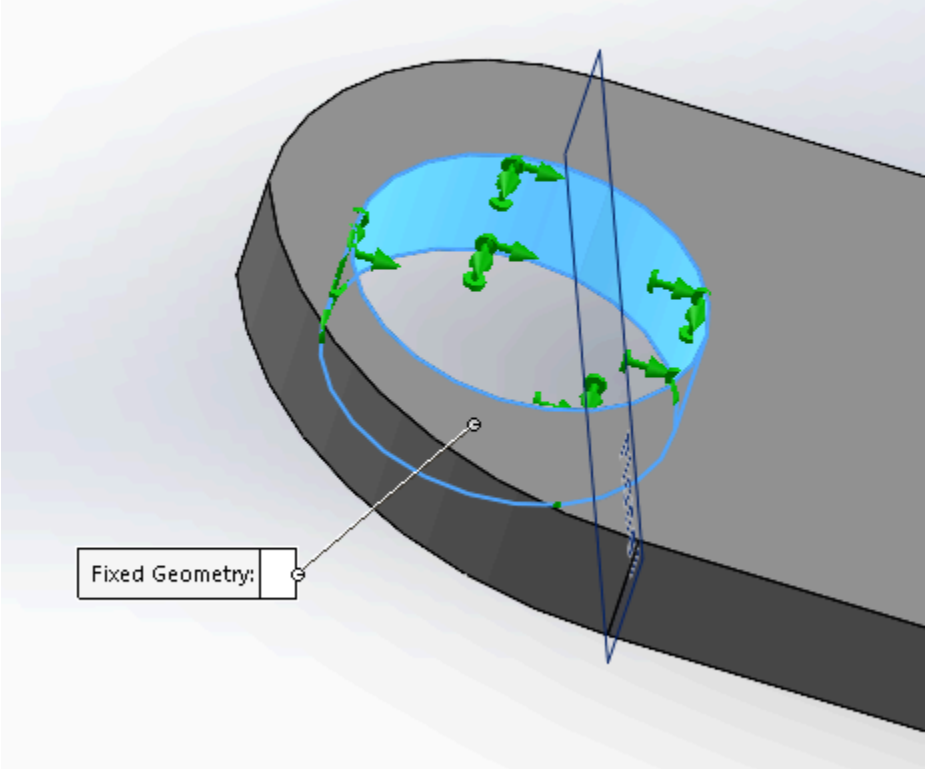
Simulation of the linkage rod is necessary because it transfers motion between the turntable handle and the upper blade turntable. While in use, the rod experiences tensile and compressive forces as the handle is actuated and the blades open and close. For the simulation, one opening of the rod was constrained to represent the pin connection to one turntable, while a force of approximately **40 N** was applied at the opposite opening to represent the load transferred from the handle mechanism. Although the rod primarily functions to transfer motion rather than support large external loads, it is important to verify that it will not fail during operation. The objective of this simulation is to confirm that the rod can withstand the transferred force without bending excessively or interfering with blade motion. The simulation results indicate that the stress levels remain below the material limits and that displacement is minimal. Therefore, the rod is expected to maintain structural integrity during normal device use.

Material: PEEK (30% glass fiber)

Fixture: fixed geometry

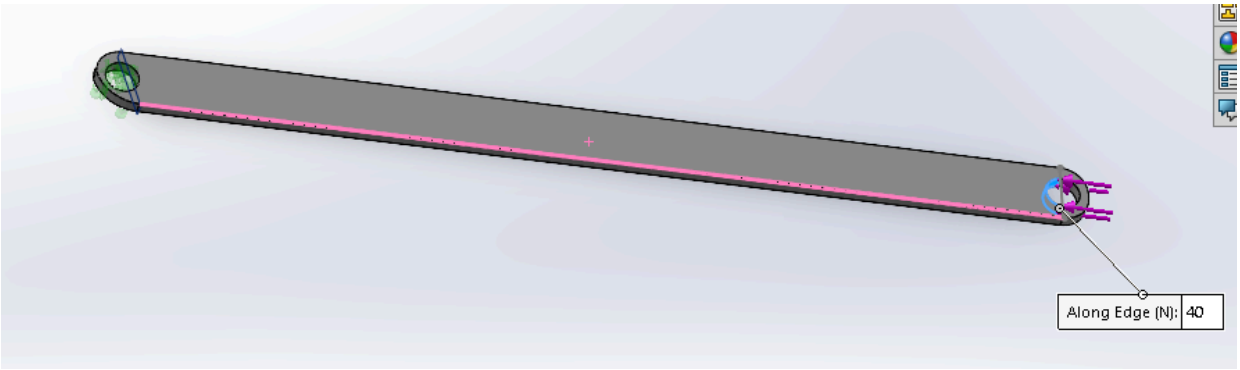


Close up image of fixed geometry:

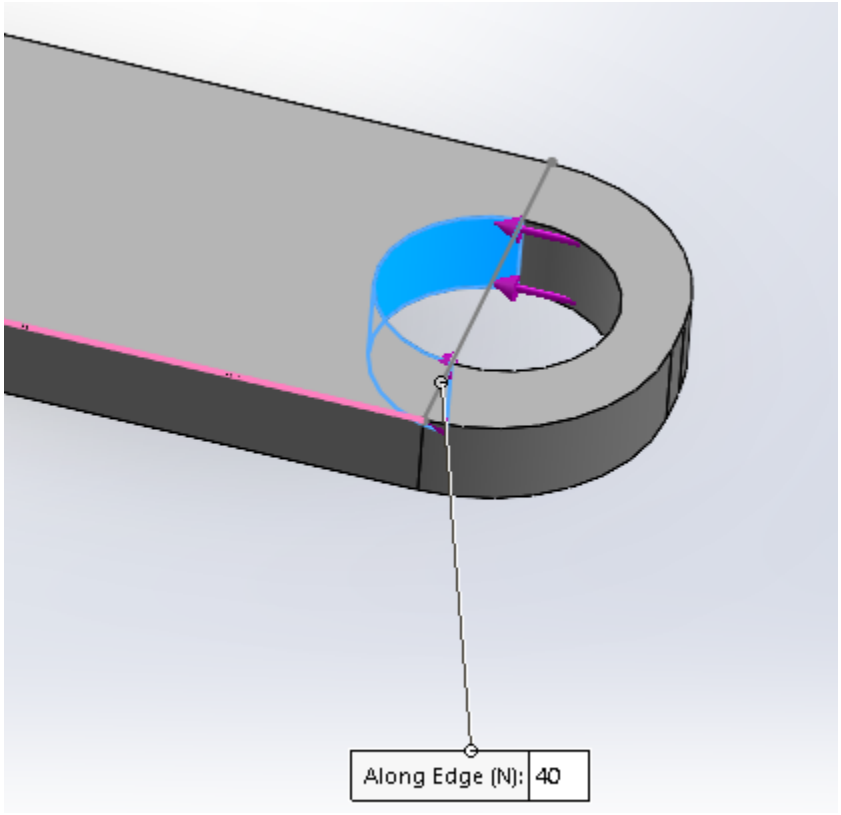


Force: 40 N (transferred load from 40N handle input)

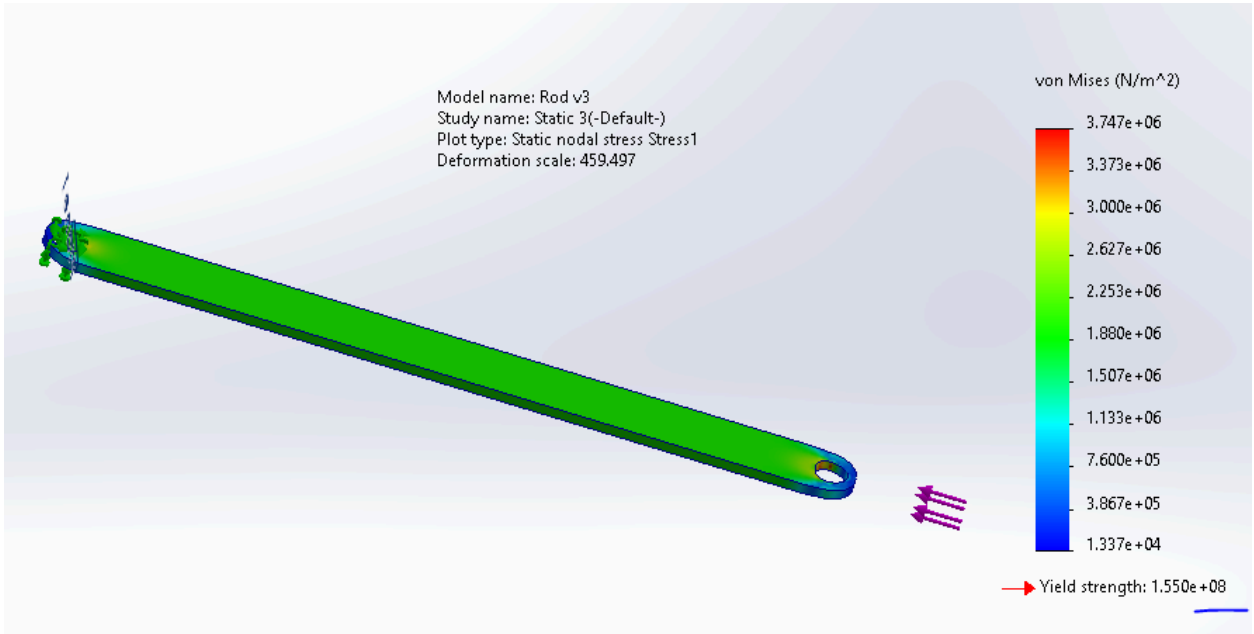
Force direction:



Close up image of force direction:



Stress plot with deformation:

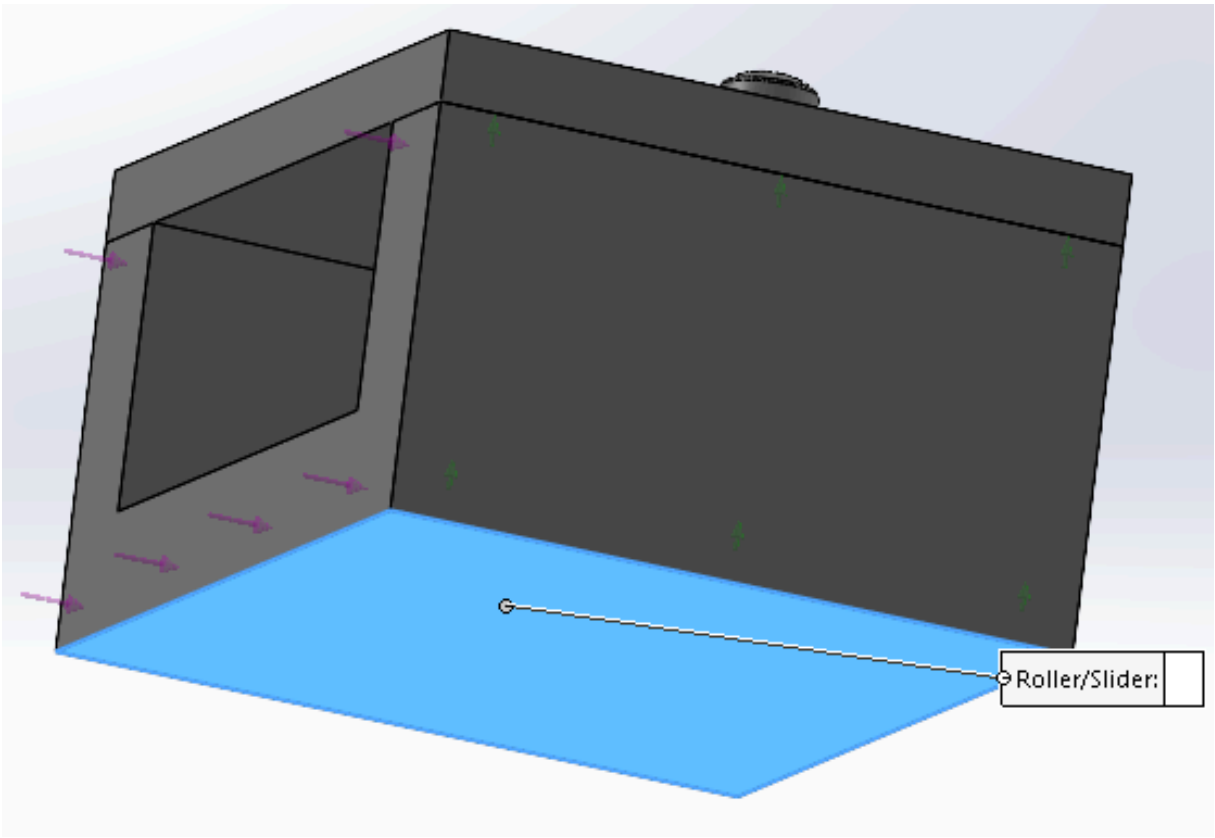


Adjuster

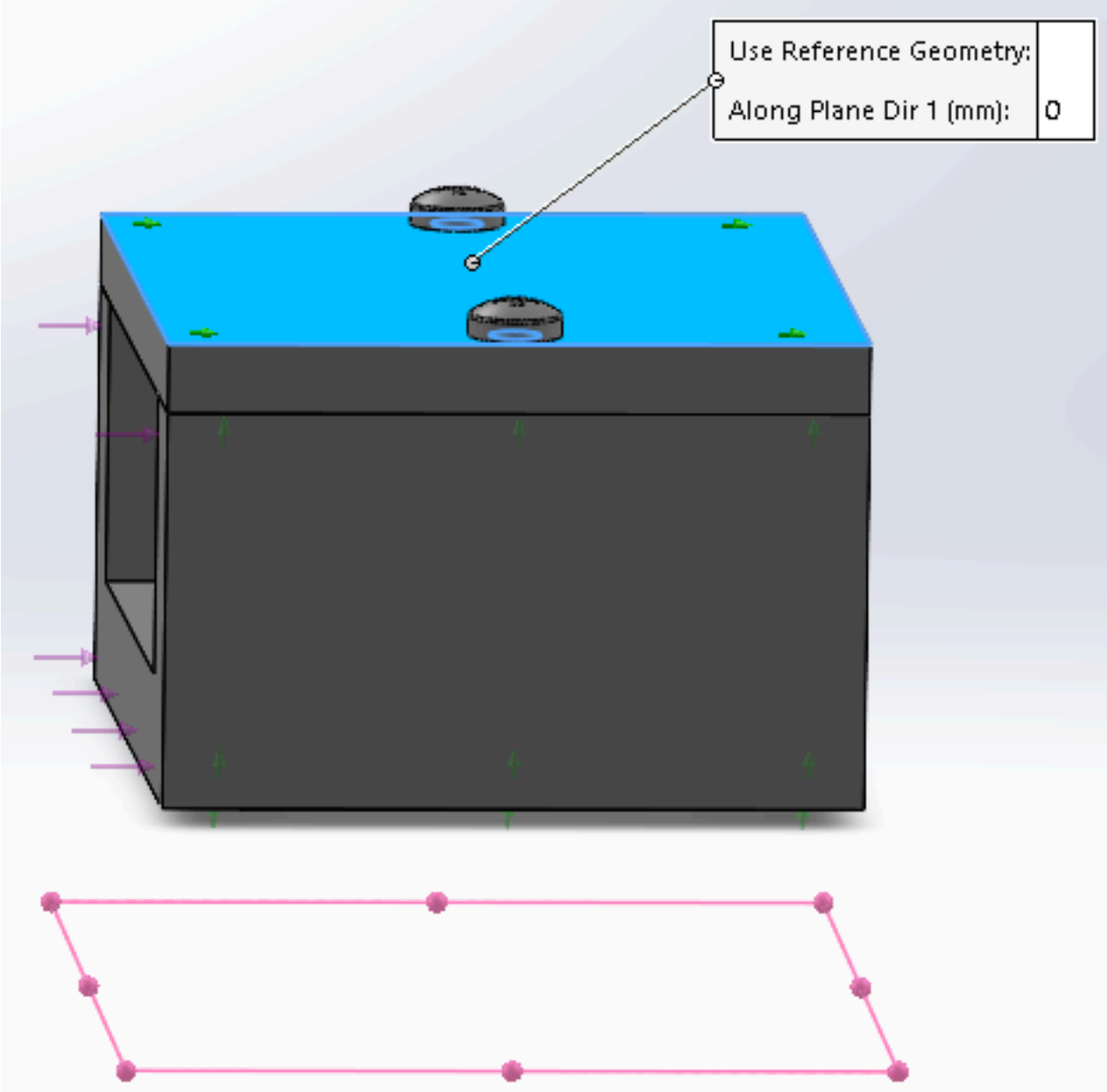
Simulation of the net adjuster is necessary because it secures the net capture mechanism and allows adjustment of net tension during use. As the blades open and close, forces from the tensioned net are transferred directly to the adjuster assembly. For the simulation, the mounting and contact regions were constrained using a combination of roller/slider and reference geometry fixtures to represent how the adjuster is supported during operation while preventing rigid-body motion in the individual part study. A force of approximately **40 N** was applied at the net attachment region to represent the load transferred through the mesh during actuation. The purpose of this analysis is to determine whether the adjuster can withstand repeated loading without excessive deformation or failure. The simulation results indicate that the component can support the expected forces while maintaining structural integrity. The displacement remains within an acceptable range, suggesting that the adjuster can continue to hold and tension the net during device use.

Material: PEEK (30% Glass Fiber)

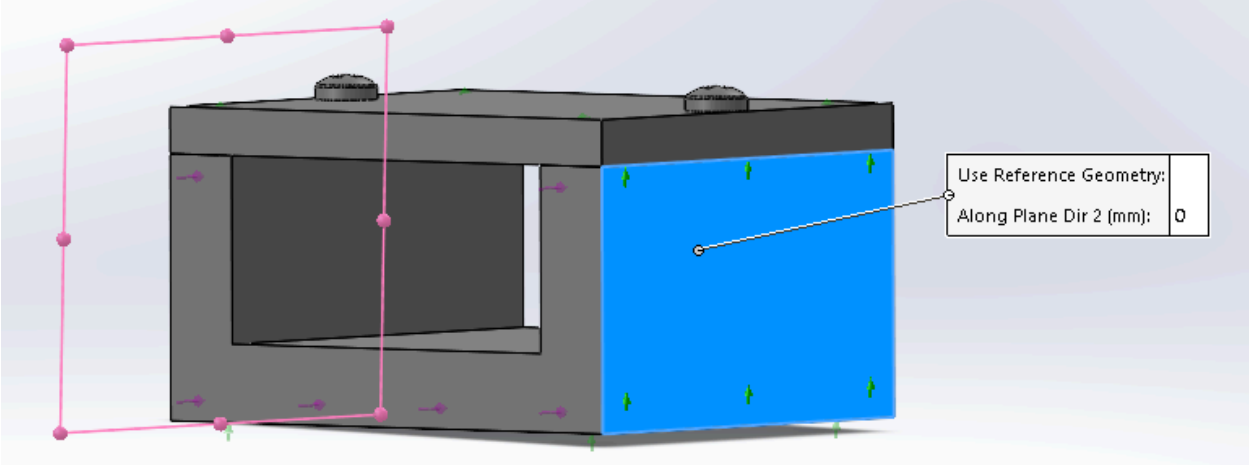
Fixture #1: Roller/Slider



Fixture #2: Reference Geometry (Top Plane)

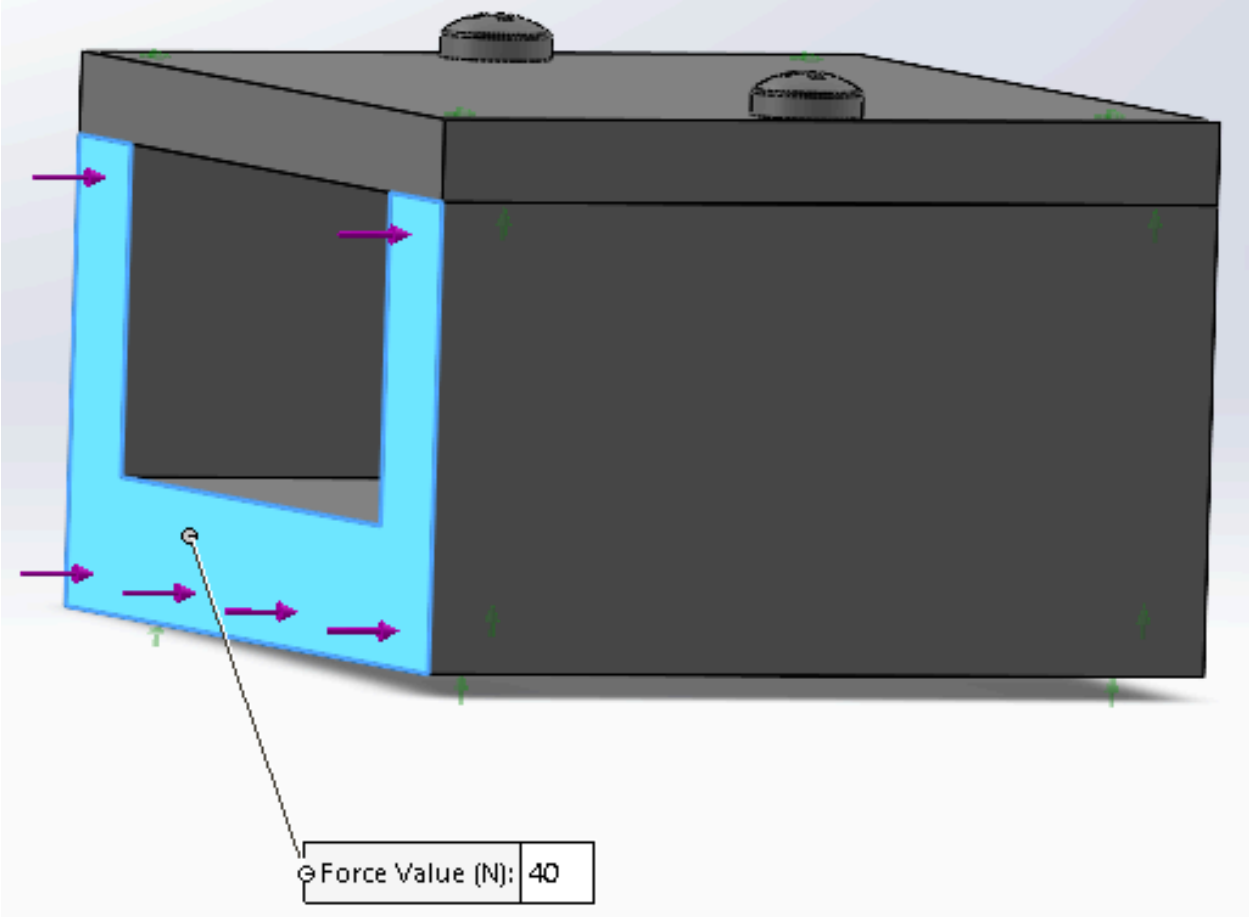


Fixture #3: Reference Geometry (Front Plane)

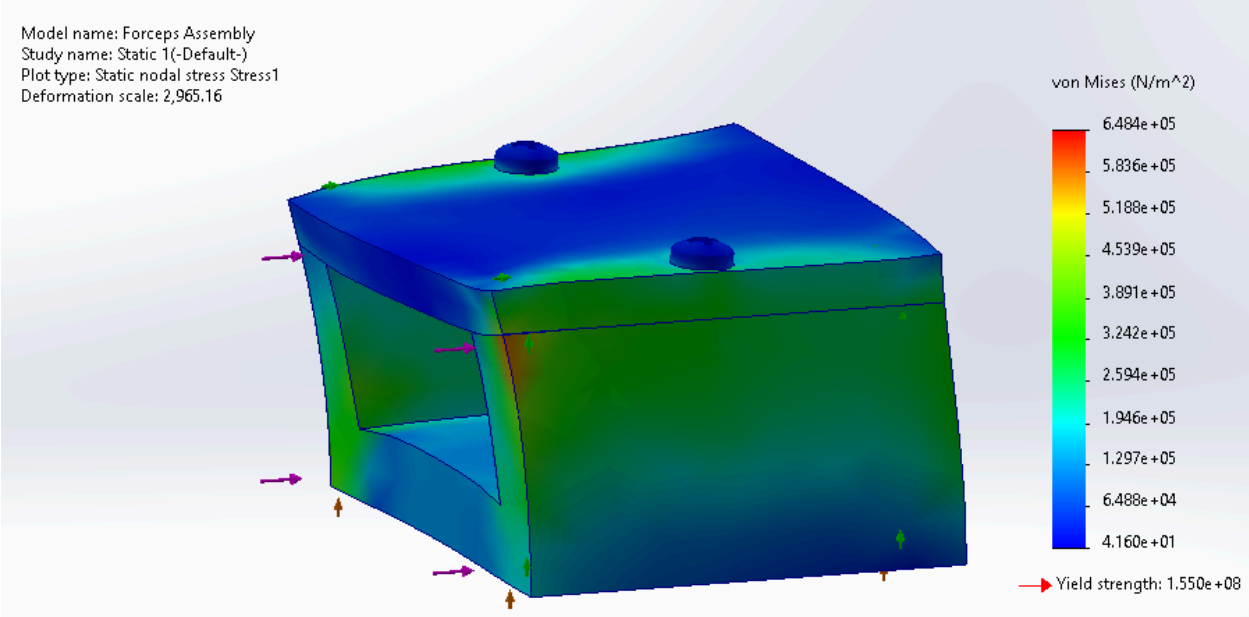


Force: 40N

Force direction:



Stress plot with deformation:



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